



JD/JW



NATIONAL CADET CORPS

UNITY & DISCIPLINE

CADET'S HAND BOOK (ARMY) SPECIALISED SUBJECT

Preface

1. National Cadet Corps (NCC), came into existence, on 15 July 1948 under an Act of Parliament. Over the years, NCC has spread its activities and values, across the length and breadth of the country; in schools and colleges, in almost all the districts of India. It has attracted millions of young boys and girls, to the very ethos espoused by its motto, "unity and discipline" and molded them into disciplined and responsible citizens of the country. NCC has attained an enviable brand value for itself, in the Young India's mind space.
2. National Cadet Corps (NCC), aims at character building and leadership, in all walks of life and promotes the spirit of patriotism and National Integration amongst the youth of the country. Towards this end, it runs a multifaceted training; varied in content, style and processes, with added emphasis on practical training, outdoor training and training as a community.
3. With the dawn of Third Millennia, there have been rapid strides in technology, information, social and economic fields, bringing in a paradigm shift in learning field too; NCC being no exception. A need was felt to change with times. NCC has introduced its New Training Philosophy, catering to all the new changes and developments, taking place in the Indian Society. It has streamlined and completely overhauled its training philosophy, objectives, syllabus, methodology etc, thus making it in sync with times. Subjects like National Integration, Personality Development and Life Skills, Social Service and Community Development activities etc, have been given prominent thrust.
4. The new syllabus, has been crystallised after obtaining a detailed feedback, from all the Directorates and the same having been brainstormed at HQ DG NCC. The syllabus has been implemented with effect from 01 May 2019.
5. For the ease of Trainees, a summary has been given at the end of each chapter. The syllabus has been revised, to make it cadet friendly, by removing the commonalities in subjects, of the school/college syllabus and making it more relevant. It is hoped, that this will facilitate, better assimilation and increased interest among the cadets.
6. The book has been the outcome, of sincere devotion and relentless effort of the Study Team ordered by HQ DG NCC. Our sincere gratitude and compliments to them. Any suggestions are welcome for its improvement in the future editions
7. Contents of this hard work, must form the basis of Institutional Training, with explicit commitment.



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Acknowledgement

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JD / JW (ARMY) SPECIALISED SUBJECTS**BLOCK SYLLABUS**

S No	Subject	Periods		
		First Year	Second Year	Total Periods
1	Armed Forces	03	03	06
2	Map Reading	09	09	18
3	Field Craft & Battle Craft	18	09	27
4	Military History	6	10	16
5	Communication	00	03	06
Total		36	34	70

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2.	AF-2	Modes of Entry into Army, Police and CAPF.	II	03	09	20
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3.	MR-1	Introduction to Map Reading	I	03	21	28
4..	MR-2	Conduct of Map Reading, Google Tourist Maps and Applications	I II	06 09	29	33
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6.	FC & BC-2	Indication of landmark	I II	03 03	36	37
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11.	MH-1	Biographies of Renowned Generals	I	03	53	54
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13.	MH-3	Study of Battles of Indo-Pak Wars 1965, 1971, & Kargil	II	03	57	58
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CHAPTER 1 : ARMED FORCES : ARMY, POLICE AND CENTRAL ARMED POLICE FORCES (CAPF)

INTRODUCTION

1. Defence Services plays a vital role in maintaining sovereignty and territorial integrity of our nation and responsibility is with Ministry of Defence and Ministry of Home Affairs. All these elements of national security strive in achieving the assigned role and task. It is an approach towards 'Unity and Integration' of nation and all cadets of NCC must understand the basic organisation of Armed Forces, Police organisations and Central Armed Police Forces of the country.

Armed Forces

ARMY

2. Indian Army is the land-based branch and the largest component of the Indian Armed Forces and The President of India is the Supreme Commander. It is headed by The Chief of Army staff (COAS), who is a four-star general

Command Headquarters

3. Command Headquarters is commanded by an officer of the rank of 'Lieutenant General', who is known as the Army Commander or General Officer Commanding – in - Chief. The whole country is divided into eight theatre Commands who have subordinate formations under them. These are: -

Command Insignia	Command Name	Headquarters
	Integrated Headquarters of Ministry of Defence (Army)	New Delhi
	Central Command	Lucknow
	Eastern Command	Kolkata
	Northern Command	Udhampur
	Southern Command	Pune
	South Western Command	Jaipur
	Western Command	Chandimandir
	Army Training Command	Shimla
	Andaman & Nicobar Command	Andaman
	Strategic Force Command	New Delhi

NAVY

4. Our country is covered almost from three sides with water with a coastline of approximately over 6000 Km. The sea around India has impact / effect on India's freedom, trade, commerce, and culture. The Chief of Naval Staff, usually a four-star officer in the rank of Admiral, commands the navy. The Indian Navy is the seventh largest in the world. The primary objective of the navy is to secure the nation's maritime borders.

Constituents of the Navy

5. As of 2018, the Indian Navy has a strength of 67,228 personnel and a large operational fleet consisting of one aircraft carrier, one amphibious transport dock, eight landing ship tanks, 11 destroyers, 14 frigates, one nuclear-powered attack submarine, one ballistic missile submarine, 13 conventionally-powered attack submarines, 23 corvettes, six mine countermeasure vessels, 29 patrol vessels, four fleet tankers and various other auxiliary vessels.

Organisation and Administration

6. Chief of Naval Staff commands Indian Navy. Integrated Headquarters of the Ministry of Defence (Navy) is located in New Delhi. The Navy is divided into three commands: -

<u>Command</u>	<u>Headquarter</u>
Western Naval Command	Mumbai
Eastern Naval Command	Vishakhapatnam
Southern Naval Command	Kochi

AIR FORCE

Air Headquarters

7. Indian Air Force is commanded by Chief of the Air Staff. The staff of Air Headquarters consists of three branches: -

- (a) Air Staff branch.
- (b) Administrative branch.
- (c) Maintenance branch.

Commands

8. The Air Force is organized into seven commands which are controlled by Air HQ. Each Command is placed under the command of an Air Officer Commanding-in-Chief. The Commands are: -

<u>Command</u>	<u>Headquarter</u>
<u>Operational commands</u>	
Central Air Command (CAC)	Allahabad, Uttar Pradesh
Eastern Air Command (EAC)	Shillong, Meghalaya
Southern Air Command (SAC)	Thiruvananthapuram, Kerala
South Western Air Command (SWAC)	Gandhinagar, Gujarat
Western Air Command (WAC)	New Delhi
<u>Functional Commands</u>	
Training Command (TC)	Bangalore, Karnataka
Maintenance Command (MC)	Nagpur, Maharashtra

POLICE ORGANISATIONS & CENTRAL ARMED POLICE FORCES

9. Central Armed Police Forces & Other Police Organisations

- (a) Border Security Force.
- (b) Central Industrial Security Force.
- (c) Central Reserve Police Force.
- (d) Indo-Tibetan Border Police.
- (e) National Security Guards.
- (f) Special Protection Group.
- (g) Sashastra Seema Bal (SSB).
- (h) Assam Rifles (AR).

10. **Central Investigation and Intelligence Institutions of Police Organisations:-**

- (a) Central Bureau of Investigation.
- (b) Indian Income-tax Department.
- (c) Directorate of Revenue Intelligence.
- (d) Central Economic Intelligence Bureau.
- (e) Directorate General of Central Excise Intelligence.
- (f) National Investigation Agency.
- (g) Narcotics Control Bureau.
- (h) Bureau of Police Research and Development (BPRD).
- (i) National Crime Records Bureau (NCRB).

11. **Central Forensic Institutions of Police Organisations:-**

- (a) Central Forensic Science Laboratory.
- (b) LNJNI National Institute of Criminology and Forensic Sciences (LNJNI-NICFS).

State Police Organisations

12. Each state and union territory of India has a state police force, headed by a director general of police ranked officer. Under the inspector general are a number of police "Ranges" composed of three to six districts, headed by Deputy Inspectors General. District police headquarters are commanded by Superintendents of Police (SP).

**BADGES OF RANKS, HONOURS AND AWARDS IN ARMED FORCES,
POLICE AND CENTRAL ARMED POLICE FORCES**

BADGES OF RANK - ARMY

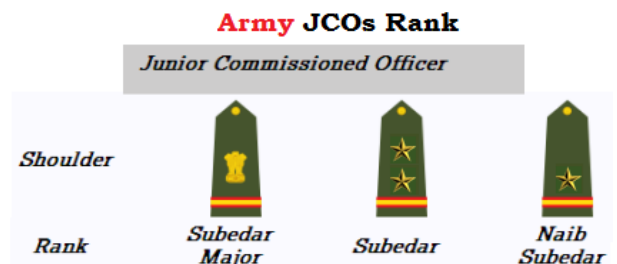
Commissioned Officers - Army

13. Commissioned Officers of Indian Army are those who command their troops from Platoon or equivalent up to armies and hold Presidents commission. Field Marshal is an honorary rank and is given to a General for his valuable services. The badges of rank worn by commissioned officers are as shown:-



Junior Commissioned Officer (JCO) Army

14. The second set of officers in the Army is Junior Commissioned Officers. The soldiers who become JCOs come up through the Non-Commissioned Officer's ranks. The badges of rank worn by the JCOs are as shown:-



Non-Commissioned Officer (NCO) Army

15. The third set of officers is the Non-Commissioned Officers (NCOs). These ranks are given to jawans according to their merit and seniority. The badges of ranks for NCOs are as shown:-



BADGES OF RANK- NAVY

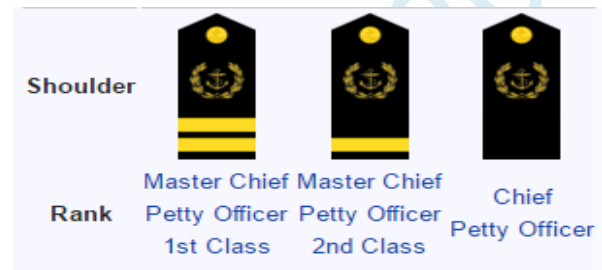
Commissioned Officers Navy

16. Admiral of the Fleet is an honorary rank given to an Admiral for his invaluable services and will continue to serve the rest of his term with the honorary rank. This rank has not been used in the Indian Navy. The badges of rank worn by Naval Officers are:-



Junior Commissioned Officers (JCOs) Navy

17. The badges of rank worn by Junior Commissioned Officers (JCOs) Navy are as shown:-



Non-Commissioned Officers (NCOs) Navy

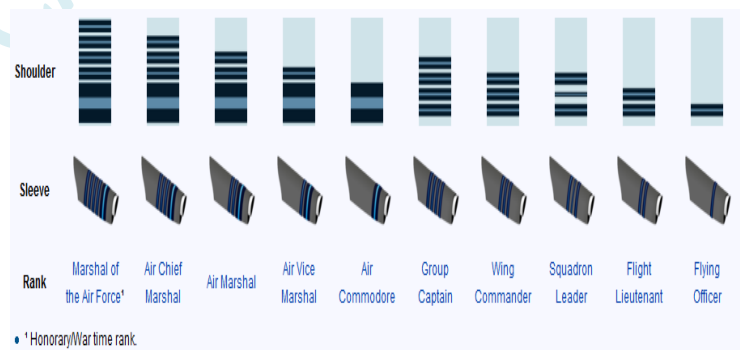
18. The badges of rank worn by the NCOs are as shown:-



BADGES OF RANK- AIR FORCE

Commissioned Officers Air Force

19. Marshal of the Air Force is an honorary rank given to an Air Chief Marshal for his invaluable services. In recognition of his services the Government of India gave the rank of Marshal of the Air Force to Arjan Singh in January 2002 making him the first and the only "Five Star" rank officer with the Indian Air Force. The badges of rank worn by officers are as shown:-



Junior Commissioned Officers (JCOs) Air Force

20. The badges of rank worn by these Officers are as shown:-



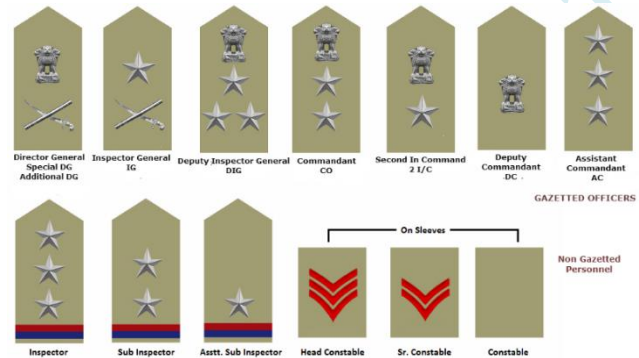
Non-Commissioned Officers (NCOs) Air Force

21. The badges of rank worn by these NCOs are as shown:-



BADGES OF RANK- POLICE & CENTRAL ARMED POLICE FORCES

22. The badges of ranks worn by CAPF & Police are as shown:-



HONOURS AND AWARDS

23. Honours and awards awarded are divided into following categories:-

- Gallantry Awards of Armed forces.
- Padma Awards.
- President's Police Medal (PPM) for Distinguished Service.
- President's Police Medal (PM) for Meritorious Service.
- Distinguished Service Awards of Armed Forces.
- Governor's Police Medals for Gallantry & Distinguished Service.
- Union Home Minister's Medal for Excellence in Investigation.
- Antrik Suraksha Seva Medal.
- Police Special Duty Medal.
- Parakram Padak.
- Kabir Puraskar
- Communal Harmony Awards
- Fire Service Medals
- Home Guard & Civil Defence Medals.

24. Gallantry Awards in the Face of Enemy (War Time).

- Param Vir Chakra.
- Maha Vir Chakra.
- Vir Chakra.
- Sena Medal / Nau Sena Medal / Vayu Sena Medal.
- Mention in Dispatch.
- Chief of Staff Commendation Card.

25. Gallantry Awards Other than in the Face of Enemy (Peace Time).

- Ashoka Chakra.
- Kirti Chakra.
- Shaurya Chakra.
- President's Police Medal (PPM) for Gallantry.
- President's Police Medal (PPM) for Distinguished Service.

26. **Non-Gallantry Awards /Distinguished Service Awards.**

- (a) Sarvottam Yudh Seva Medal.
- (b) Param Vishisht Seva Medal.
- (c) Uttam Yudh Seva Medal.
- (d) Ati Vishisht Seva Medal.
- (e) Yuddh Seva Medal.
- (f) Vishisht Seva Medal.

ROLE OF ARMY, POLICE & CAPF

Command and Control

27. The Integrated Headquarters of Ministry of Defense (Army) functions under the Ministry of Defence and is located at New Delhi. Command HQs have field formations and static formations under them.

Field Formations

28. These are the field forces for the battle, grouped into following three categories:-

- (a) Corps.
- (b) Divisions.
- (c) Brigades.

Static Formations

29. Area Headquarters, Sub Area Headquarters and Station Headquarters are commanded by an officer of the rank of Lieutenant General, Major General and Brigadier respectively. These formations are extended all over the country and look after the infrastructural assets, lines of communication logistics, administration and all civil-military matters.

FIGHTING ARMS, ROLE

30. **Armoured Corps.** To destroy the enemy by relentless mobile and offensive action in both offensive as well as defensive operations.

31. **Infantry.** Infantry is the main combat arm, best arm suited for close battle. Its role is to close in with the enemy and destroy him or capture his territory. In Defence, they hold the ground against all types of attack. They also take part in counter insurgency and counter terrorist operations. Infantry also provides assistance in aid to civil authorities when called for.

32. **Mechanised Infantry.** It is infantry with more mobility and fire power. Mechanized Infantry operates in armoured personnel carrier (APC) which has enough protection against small arms fire. They operate along with Armour in battle field. Equipments used are BMP- I and II.

SUPPORTING ARMS, ROLE

33. **Artillery.** The role of Artillery is to provide such fire power in the battle area that enemy neither interferes with our operations, nor develops his own effectively.

34. **Army Air Defence.** Army Air Defence is equipped with air defence guns and Short and Medium range surface to air missile systems. Along with Air force it provides air defence to mobile forces, Vulnerable Areas and Vulnerable Points.

35. **Army Aviation Corps.** The role of Army Aviation corps is to carry out recce and observation and also to provide commanders and staff, rapid means of communication for liaison visits and reconnaissance. It is ideally suited for evacuation of battle casualties.

36. **Corps of Engineers.** It is one of the oldest supporting arms of the Indian army. The role of Engineers in War is to provide support for offensive and defensive operations in mine warfare, bridging, demolition, constructions of field fortifications and operational roads/tracks.

37. **Corps of Signals.** They are the communicators of Indian Army and provide Radio, Radio Relay, data and line communication and establish Signal Centres during war and peace. It also monitors enemy's communication systems.

38. **Intelligence Corps.** The main task of Intelligence Corps is to gather intelligence of the enemy and prevent the leakage of own information to the enemy.

SUPPORTING SERVICES

39. The following are Supporting Services, as they provide logistical support to fighting & Supporting Arms:-

- (a) Corps of Electrical and Mechanical Engineering
- (b) Army Service Corps
- (c) Army Medical Corps
- (d) Army Dental Corps
- (e) Military Nursing Service
- (f) Army Ordnance Corps
- (g) Remount and Veterinary Corps
- (h) Military Farms
- (i) Corps of Military Police
- (j) Judge Advocate General Branch
- (k) Military Engineering Services
- (l) Border Roads Organisation
- (m) Army Pioneer Corps

ROLE OF CENTRAL ARMED POLICE FORCES & POLICE

Central Armed Police Forces & Other Police Agencies

40. **Assam Rifles (AR).** Assam Rifles (AR) is the oldest paramilitary force in India whose primary role is to protect part of India's North eastern borders and also the internal security also. It also maintained law and order in the tribal areas.

41. **Border Security Force.** The Indian Border Security Force (BSF) is responsible for policing India's land borders during peacetime and preventing trans-border crimes along Pakistan and Bangladesh.

42. **Central Industrial Security Force.** The primary task of CISF is providing industrial security. The Central Industrial Security Force (CISF) guards industrial installations around the country owned by the Central government as well as securing seaports and airports. The CISF also provides security to certain NGOs. They provide security for atomic power plants, space installations, mines, oil fields and refineries, major ports, heavy engineering plants, steel plants, barrages, fertilizer units, airports, hydroelectric/thermal power plants and other installations partially or wholly run by the government.

43. **Central Reserve Police Force.** The Central Reserve Police Force (CRPF) is one of the largest Central Police organizations in the world. Its main objective is to assist and help states and union territories' law enforcement agencies in maintaining law and order and to contain insurgency

44. **Indo-Tibetan Border Police.** The Indo-Tibetan Border Police (ITBP) is responsible for security along the Indo-Tibetan Border.

45. **National Security Guards.** It is India's premier counter-terror outfit and is typically deployed in situations that would be beyond the capabilities of regular police units. An NSG team with a dedicated transport aircraft is always stationed at Palam airport in New Delhi, ready to deploy in 30 minutes. The NSG has also been increasingly tasked with the protection of VIPs.

46. **Special Protection Group.** The Special Protection Group (SPG) is the executive protection agency of the Government of India. It provides the security 24 by 7 all over India to Prime Minister including ex-prime Ministers and their Family Members at any location across India.

47. **Sashastra Seema Bal (SSB).** The Sashastra Seema Bal, formed in the year 1963, deployed at the Indo-Nepal and Indo-Bhutan borders. The SSB personnel are trained in the fields of law and order, military tactics, jungle warfare, counter-insurgency, and internal security.

48. **Other Central investigation and intelligence agencies are as under: -**

- (a) Central Bureau of Investigation.
- (b) Indian Income-tax Department.
- (c) Directorate of Revenue Intelligence.
- (d) Central Economic Intelligence Bureau.
- (e) Directorate General of Central Excise Intelligence.
- (f) National Investigation Agency.
- (g) Narcotics Control Bureau.
- (h) Bureau of Police Research and Development (BPRD).
- (i) National Crime Records Bureau (NCRB).
- (j) Central Forensic Science Laboratory.
- (k) National Institute of Criminology and Forensic Sciences.

CONCLUSION

49. The organisation of the Armed Forces, Police organisation and Central Armed Police Forces is structured in a manner to facilitate coordination of the functioning of all the services with the nucleus being the Headquarters and formations. As NCC cadets and responsible youth it is essential to understand these organisations, which play major role in national security, development as well as in nation building.

NATIONAL CADET CORPS

CHAPTER 2 : MODES OF ENTRY TO ARMY, CAPF AND POLICE

Introduction

1. The Indian Army's primary mission is to ensure the national security and defence of the Republic of India from external threats and maintaining peace and security. It also provides aid to civil authorities in maintaining essential services. It is important for NCC cadets to understand how to join services and police organisations.

ARMY

2. The Army offers both Permanent and Short Service Commission. Permanent Commission (PC) is granted through the Indian Military Academy (IMA) Dehradun and Short Service Commission (SSC) is granted through Officers Training Academy (OTA) Chennai:-

(a) **Permanent Commission.**

- (i) NDA / AFMC - after 10+2 (Through UPSC).
- (ii) Direct Entry (Through UPSC).
- (iii) Engineering Graduates – TGC.
- (iv) University Entry Scheme.
- (v) Technical Entry Scheme 10 +2.
- (vi) Army Cadet College Wing, after 10+2.
- (vii) SCO / PCSL Entry from Regular Army.

(b) **Short Service Commission.**

- (i) Non-Tech (Both Men & Women).
- (ii) Tech (Both Men & Women).
- (iii) NCC Special Entry (Both Men & Women).
- (iv) Law Graduates (Both Men & Women).

Note. Details of eligibility criteria, duration and venue of training and other information is freely available through newspapers or may be obtained from www.joinindianarmy.nic.in.

ENTRY SCHEMES OFFICERS (MEN AND WOMEN)

3. **National Defence Academy (NDA).**

Vacancies Per Course	320 (Twice a year) Army – 208, Air Force – 70, Navy - 42
Notification Published in Employment News and leading Daily news Paper	Jun and Dec, Notified by UPSC
Eligibility Criteria	
Age	16 ½ to 19½ yrs as on first day of the month in which course is due to commence
Qualification	12 th Class of 10+2 System of Education of Equivalent for Army and with Physics and Maths for AF/Navy
Marital Status	Un Married
How to Apply	Apply Online on UPSC Website as per dates given in the Notification in the months of Jun and Dec.
Likely SSB Date	Sep to Oct for Jan Course and Jan to Apr for Jul Course
Date Commencement of Training	Jan and Jul
Training Academy	NDA , Khadakwasla, Pune
Duration of Training	3 Yrs at NDA and 1 Yr at IMA

4. **10+2 Technical Entry Scheme (TES).**

Vacancies Per Course	90 (Twice a year)
Notification Published in Employment News and leading Daily news Paper	Notified by DG Rtg AG's branch IHQ of MOD(Army) in May/Jun & Oct/Nov
Eligibility Criteria	
Age	16 ½ to 19½ yrs as on first day of the month in which course is due to commence
Qualification	10+2 Physics, Chemistry & Maths (aggregate 70% and above to apply)
Marital Status	Un Married
How to Apply	Apply online on Official website of DG Rtg www.joinindianarmy.nic.in as Notified in the Notification.
Likely SSB Date	Aug to Oct and Feb to Apr
Date Commencement of Training	Jan and Jul
Training Academy	OTA Gaya
Duration of Training	5 Yrs (1 Yr OTA GAYA & 4 Yrs CTWS) Permanent Commission after 4 Yrs. Phase-I- 01 Year Pre-commissioning Training at OTA Gaya & 03 Years at CME Pune/MCTE Mhow/MCEME Secunderabad Phase-II- 01 Year Post commission training at CME Pune/MCTE Mhow/MCEME Secunderabad

5. **Indian Military Academy Direct (IMA).**

Vacancies Per Course	200 (Twice a year).
Notification Published in Employment News and leading Daily news Paper	Notified by UPSC under the aegis of CDSE in Jul and Nov
Eligibility Criteria	
Age	19 to 24 years
Qualification	Graduation from Recognised University.
Marital Status	Un Married
How to Apply	Apply Online on UPSC Website as per dates given in the Notification in the months of Jul and Nov.
Likely SSB Date	Aug / Sep and Feb / Mar
Date Commencement of Training	Jan and Jul
Training Academy	IMA, Dehradun
Duration of Training	1 ½ Years

6. **Officers Training Academy (Non-Technical Men).**

Vacancies Per Course	175 (Twice a year)
Notification Published in Employment News and leading Daily news Paper	Notified by UPSC under the aegis of CDSE in Jul and Nov
Eligibility Criteria	
Age	19 to 25 yrs
Qualification	Graduation from Recognised University.
Marital Status	Un Married
How to Apply	Apply Online on UPSC Website as per dates given in the Notification in the months of Jul and Nov.
Likely SSB Date	Nov / Dec and May / Jun
Date Commencement of Training	Oct and Apr
Training Academy	OTA Chennai
Duration of Training	49 Weeks

7. **Judge Advocate General (JAG) Men.**

Vacancies Per Course	10(As Notified)(Twice a year)
Notification Published in Employment News and leading Daily news Paper	Notified by Directorate General Recruiting / AG Branch in Jul/Aug and Jan/Feb
Eligibility Criteria	
Age	21 to 27 years
Qualification	Graduate with LLB with 55% marks. Eligible for Registration with Bar Council of India / State.
Marital Status	Un Married
How to Apply	Apply online on Official website of DG Rtg www.joinindianarmy.nic.in as Notified in the Notification.
Likely SSB Date	Dec / Jan and Jul / Aug
Date Commencement of Training	Apr and Oct
Training Academy	OTA, Chennai
Duration of Training	49 Weeks

8. **NCC (Spl) Entry Men.**

Vacancies Per Course	50 (Twice a Year)
Notification Published in Employment News and leading Daily news Paper	Notified by Directorate General Recruiting / AG Branch in Jun and Dec
Eligibility Criteria	
Age	19 to 25 years
Qualification	Graduate with 50% Aggregate marks, Two Years' service in NCC Senior Div Army with minimum 'B' Grade in 'C' Certificate Exam.

Marital Status	Un Married
How to Apply	Apply offline as Notified in the Notification through NCC Dte
Likely SSB Date	Dec / Jan and Jun / July
Date Commencement of Training	Apr and Oct
Training Academy	OTA, Chennai
Duration of Training	49 Weeks

9. **University Entry Scheme (UES).**

Vacancies Per Course	60 (As Notified) (Once a Year)
Notification Published in Employment News and leading Daily news Paper	Notified by Directorate General Recruiting / AG Branch in Jun/Jul
Eligibility Criteria	
Age	18 to 24 years
Qualification	Pre Final year students of Notified Engineering Streams
Marital Status	Un Married
How to Apply	Apply online on Official website of DG Rtg www.joinindianarmy.nic.in as Notified in the Notification.
Likely SSB Date	Nov to Feb
Date Commencement of Training	Jul
Training Academy	IMA, Dehradun
Duration of Training	One Year

10. **Technical Graduate Course (TGC) Engineers.**

Vacancies Per Course	60 (As Notified) (Twice a Year)
Notification Published in Employment News and leading Daily news Paper	Notified by Directorate General Recruiting / AG Branch in Mar/Apr and Sep/Oct.
Eligibility Criteria	
Age	20 to 27 years
Qualification	BE / B Tech in notified streams of Engineering
Marital Status	Un Married
How to Apply	Apply online on Official website of DG Rtg www.joinindianarmy.nic.in as Notified in the Notification.
Likely SSB Date	Mar / Apr and Sep / Oct
Date Commencement of Training	Jan and Jul
Training Academy	IMA, Dehradun
Duration of Training	One Year

11. **Short Service Commission Tech (Men).**

Vacancies Per Course	100 (As Notified) (Twice a year)
Notification Published in Employment News and leading Daily news Paper	Notified by Directorate General Recruiting / AG Branch in Jun / Jul and Dec/Jan
Eligibility Criteria	
Age between	20 to 27 years
Qualification	Engineering Degree in notified stream
Marital Status	Un Married
How to Apply	Apply online on Official website of DG Rtg www.joinindianarmy.nic.in as Notified in the Notification.
Likely SSB Date	Dec-Jan and Jun-Jul
Date Commencement of Training	Apr and Oct
Training Academy	OTA, Chennai
Duration of Training	49 Weeks

12. **Army Education Corps (AEC) Men.**

Vacancies Per Course	20 (As Notified) (Twice a Year)
Notification Published in Employment News and leading Daily news Paper	Notified by Directorate General Recruiting / AG Branch in Mar/Apr and Sep/Oct
Eligibility Criteria	
Age	23 to 27 years
Qualification	MA / MSc in 1st or 2nd division in notified subjects from recognised university.
Marital Status	Married and Un Married
How to Apply	Apply online on Official website of DG Rtg www.joinindianarmy.nic.in as Notified in the Notification.
Likely SSB Date	Mar / Apr and Sep / Oct
Date Commencement of Training	Jan and Jul
Training Academy	IMA, Dehradun
Duration of Training	One Year

13. **Army Cadet College (ACC).**

Vacancies Per Course	75 (Twice in a Year)
Tentative Month of Publication of Notification	Notified by MP Directorate / AGs Br through Units in Mar and Aug
Eligibility Criteria	
Age	20 to 27 years , minimum service of two years one can apply
Qualification	10+2 Pattern Class with 12th pass or Equivalent. Should have qualified ACC written test
Marital Status	Unmarried / Married
Likely Date of SSB	Sep to Nov for Jan Course Mar to May for Jul Course
Training Academy	ACC Wing at IMA, Dehradun

Duration of Training	Four Years (Three Years in ACC Wing & One Year at IMA)
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14. **Permanent Commission (PC SL).**

Vacancies Per Course	100 (Once in a Year)
Tentative Month of Publication of Notification	Notified by MP Dte / AGs Br through Units in Apr and Jul
Eligibility Criteria	
Age	Maximum age 42 years, 45 years for Engineers & 45 years for IOB Minimum service 10 years
Qualification	Matric & Above
Marital Status	Unmarried / Married Serving JCOs / NCOs excluding Sepoys
Likely Date of SSB	Sep and Oct for Jul Course
Training Academy	AEC Centre & College, Pachmarhi & IMA, Dehradun
Duration of Training	Eight Weeks at AEC Centre & College, Pachmarhi & Four Weeks at IMA

15. **Special Commissioned Officer (SCO).**

Vacancies Per Course	100 (Twice in a Year)
Tentative Month of Publication of Notification	Notified by MP Directorate / AGs Br through Units in Apr and Jul
Eligibility Criteria	
Age	Between 30 to 35 years. minimum service of five years can apply
Qualification	Matric with One Year Diploma or above
Marital Status	Unmarried / Married
Likely Date of SSB	Jul/Aug for Jan Course & Nov/Dec for Jul Course
Training Academy	OTA, Gaya
Duration of Training	Eight Weeks at AEC Centre & College, Pachmarhi & One Year at OTA, Gaya

16. **Territorial Army (TA).**

Vacancies Per Course	Notified by Line Directorate (Twice in a Year)
Tentative Month of Publication of Notification	Notified by TA Directorate
Eligibility Criteria	
Age	18 to 42 years
Qualification	Graduate from a recognised university
Marital Status	Unmarried / Married
Likely Date of SSB	Aug / Sep & May / Jun
Training	Recruit Training. 30 days under TA Battalion. Annual Training Camp. Two calendar month in each training year.

	Post Commission Training. Three month in IMA within two years of commissioning.
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17. **Remount & Veterinary Corps (RVC).**

Vacancies Per Course	Notified by Line Directorate
Tentative Month of Publication of Notification	Nov / Dec , Notified by RVC Directorate
Eligibility Criteria	
Age	21- 32 years
Qualification	BVSc & AH
Marital Status	Unmarried / Married
Likely Date of SSB	Sep / Oct
Training Academy	RVC, Meerut
Duration of Training	Three months

18. **Army Medical Corps (Non-Technical).**

Vacancies Per Course	Notified by Line Directorate
Tentative Month of Publication of Notification	Notified by AMC Directorate
Eligibility Criteria	
Age	Max 42 years
Qualification	Minimum Matric
Marital Status	Unmarried / Married
Likely Date of SSB	Nov / Dec
Training Academy	AMC Centre & College.
Duration of Training	Three months

19. **Army Postal Service (APS).**

Vacancies Per Course	Notified by Line Directorate
Tentative Month of Publication of Notification	Notified by APS Directorate
Eligibility Criteria	
Age	Max 45 years
Qualification	Minimum Matric
Marital Status	Unmarried / Married
Likely Date of SSB	Sep /Oct

WOMEN ENTRY SCHEMES

20. **Short Service Commission (Non-Technical) Women.**

Vacancies Per Course	12 (Twice a year)
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Notification Published in Employment News and leading Daily news Paper	Notified by UPSC under the aegis of CDSE in July and Nov
Eligibility Criteria	
Age	19 to 25 years
Qualification	Graduation from Recognized University.
Marital Status	Un Married
How to Apply	Apply Online on UPSC Website as per dates given in the Notification in the months of Jul and Nov.
Likely SSB Date	Jun / July and Nov / Dec
Date Commencement of Training	Oct and Apr
Training Academy	OTA Chennai
Duration of Training	49 Weeks

21. **NCC Special Entry Scheme (Women).**

Vacancies Per Course	Four (Twice a year)
Notification Published in Employment News and leading Daily news Paper	Notified by Directorate General Recruiting / AG Branch in Jun and Dec
Eligibility Criteria	
Age	19 to 25 years
Qualification	Graduate with 50% Aggregate marks, Two Years service in NCC Senior Div Army with minimum 'B' Grade in 'C' Certificate Exam.
Marital Status	Un Married
How to Apply	Apply online as Notified in the Notification through NCC Dte.
Likely SSB Date	Dec / Jan and Jun / July
Date Commencement of Training	Apr and Oct
Training Academy	OTA, Chennai
Duration of Training	49 Weeks

22. **Short Service Commission for Women (SSCW) JAG.**

Vacancies Per Course	Four (As Notified) (Twice a year)
Notification Published in Employment News and leading Daily news Paper	Notified by Directorate General Recruiting / AG Branch in Jul/Aug and Jan/Feb.
Eligibility Criteria	
Age	21 to 27 years
Qualification	Graduate with LLB with 55% marks. Eligible for Registration with Bar Council of India / State.
Marital Status	Un Married
How to Apply	Apply online on Official website of DG Rtg www.joinindianarmy.nic.in as Notified in the Notification.
Likely SSB Date	Dec / Jan and Jun / Jul
Date Commencement of Training	Apr and Oct

Training Academy	OTA, Chennai
Duration of Training	49 Weeks

23. **Short Service Commission (Technical) Women.**

Vacancies Per Course	20 (As Notified) (Twice a year)
Notification Published in Employment News and leading Daily news Paper	Notified by Directorate General Recruiting / AG Branch in Jun / Jul and Dec/Jan
Eligibility Criteria	
Age	20 to 27 years
Qualification	Engineering Degree in notified stream.
Marital Status	Un Married
How to Apply	Apply online on Official website of DG Rtg www.joinindianarmy.nic.in as Notified in the Notification.
Likely SSB Date	Dec-Jan and Jun-July.
Date Commencement of Training	Apr & Oct
Training Academy	OTA Chennai
Duration of Training	49 Weeks

JCOS & OTHER RANKS

24. **Educational Qualification and Age Criteria.**

Ser. No	Category	Education	Age
(1)	Soldier (General Duty) (All Arms)	CI 10th /Matric pass with Min 45% mks in aggregate and Min 33% in each subject.	17 ½ - 21 Yrs
(2)	Soldier Technical (Technical Arms, Artillery, Army Air Defence)	(a) Sol Tech 10+2/Intermediate Exam pass in Science with Physics, Chemistry, Maths and English with min 50% marks in aggregate and min 40% marks in each subject. (b) Sol Tech (Aviation & Ammunition Examiner) 10+2/Intermediate Exam pass in Science with Physics, Chemistry, Maths and English with min 50% marks in aggregate and min 40% marks in each subject.	17 ½ - 23 Yrs
(3)	Soldier Clerk / Store Keeper Technical (All Arms)	10+2/Intermediate Examination Pass in any stream (Arts, Commerce, Science) with min 60% marks in aggregate and min 50% in each subject. Securing 50% in English and Maths/Accounts/Book keeping in Class 12th is mandatory.	17 ½ - 23 Yrs
(4)	Soldier Nursing Assistant (Army Medical Corps)	10+2/Intermediate Exam pass in Science with Physics, Chemistry, Biology and English with min 50% marks in aggregate and min 40% in each subject.	17 ½ - 23 Yrs

(5)	Sepoy Pharma (Army Medical Corps)	10+2 or equivalent exam passed with Physics, Chemistry, Biology and English, qualified in D Pharma with minimum 55% marks in aggregate and registered with State Pharmacy Council/Pharmacy Council of India. Individual qualified in B Pharma with minimum 50% marks and registered with State Pharmacy Council/Pharmacy Council of India will also be eligible.	19 - 25 Yrs
(6)	Soldier Nursing Assistant Veterinary (Remount Veterinary Corps)	10+2/Intermediate Exam pass in Science with Physics, Chemistry, Biology and English with min 50% marks in aggregate and min 40% in each subject.	17 ½ - 23 Yrs
(7)	Soldier Tradesmen (All Arms except Syce, Mess Keeper and House Keeper)	i) 10 th Simple Pass. ii) No stipulation in aggregate percentage but should have scored min 33% in each subject.	17 ½ - 23 Yrs
(8)	Soldier Tradesmen (Syce, Mess Keeper and House Keeper)	i) 8 th Simple Pass. ii) No stipulation in aggregate percentage but should have scored min 33% in each subject.	17 ½ - 23 Yrs
(9)	Survey Automated Cartographer (Engineers)	BA/BSc with Maths. Must have also passed 12 th class (10+2) or equivalent with Maths and Science as main subjects.	20-25 Yrs
(10)	Junior Commissioned Officer Religious Teacher (All Arms)	As per the Notification.	25-34 Yrs
(11)	Junior Commissioned Officer Catering (Army Service Corps)	Diploma in Hotel Management and Catering Technology.	21-27 Yrs
(12)	Havildar Education (Army Education Corps)	Group X - MA / MSc / MCA or BA / BSc / BCA / BSc (IT) with B Ed Group Y - BSc/BA/BCA/B Sc (IT) (w/o B Ed.)	20-25 Yrs
(13)	Soldier (General Duty) Women Military police	Minimum education qualification is Matric /10th/SSLC or equivalent with 45% marks in aggregate and minimum 33% marks in each subject studied at matric/10th/SSLC level, without mentioning any specific subjects. (Note: # Upper Age limit will be relaxable upto 30 yrs of age (as on date of joining training) in respect of widows of Defence personnel who have died in harness.)	17 ½ - 21 Yrs #

POLICE ORGANISATIONS AND OTHER CENTRAL ARMED POLICE FORCES

Entry into Police Services as an Officer

25. There are two ways of getting into Indian Police Services: -

- Passing All India Civil Services Examination conducted by All India UPSC Civil Services Examinations.
- Promotions from State Cadre Police Services.

26. **Central Armed Police Forces Officer Cadre and Personnel.** The Central Armed Police Forces have their own officers and personnel to these forces are recruited directly as well as obtained from respective departments. Recruitment is conducted mainly in three modes.

- Gazetted Officers.
- Subordinate Officers.

(c) Constables.

27. **Assistant Commandant.** Out of total vacancies of Assistant Commandants in CAPF, 50% are filled by Direct Recruitment (including 10% from SSCO), 33% by promotion from Inspector to AC & 17% by limited Departmental Competitive examination be conducted by CAPFs.

28. **Eligibility Criteria for Entries into Police & CAPF.**

Rank	Age	Educational Qualification	Enrolment Criteria	Relaxation	Physical Standards	
Inspector	20 to 25 years	Degree of a recognized University or equivalent	33% of SI (GD) are filled by Direct recruitment (Including 10% from ESM), 50% by promotion, 17% by limited Departmental	Relaxable for SC/ST/OBC in accordance with the instructions issued by the Govt from time to time.	Height -170 Cms	Relaxable by 5 cms for candidates from Hill areas
					Chest - 80-85 Cms	77-82 cms for STs
					Weight - Proportionate to height	-do-
Assistant Sub Inspector Radio Mechanic	18 to 25 years	Matric with 3-year Diploma in Radio and TV technology / Electronics / Tele-Communications / Computer / Electrical / Mechanical / Engineering / Domestic appliance from an Institution recognized by State/Central Government or 10+2		Relaxable for SC/ST/OBC in accordance with the instructions issued by the Govt from time to time.	Height Gen-170 Cms	Tribal/Adivasi-162.5 Cms
					Chest - 80-85 Cms	77-82Cms for STs
					Weight - Proportionate to height	-do-
Head Constable (Operator)	18 to 23 years	Matric or equivalent plus two years ITI certificate in Radio and TV/ Electronics or Intermediate or 10+2 or equivalent with Physics, Chemistry and Mathematics from a recognized Board of Education		Relaxable for SC/ST/OBC in accordance with the instructions issued by the Govt from time to time.	Height -170 Cms	Relaxable by 5 Cms for candidates from Hill areas
					Chest - 80-85 Cms	77-82Cms for STs
					Weight - Proportionate to height	-do-
Head Constable (Fitter)	18 to 23 years	Matric or equivalent plus two years ITI certificate in engine/fitter/Diesel Mechanic/Automobile/Motor Mechanic from an Institution recognized by State/Central Government or equivalent or 10+2		Relaxable for SC/ST/OBC in accordance with the instructions issued by the Govt from time to time.	Height -170 Cms	Relaxable by 5 Cms for candidates from Hill areas
					Chest - 80-85 Cms	77-82Cms for STs
					Weight - Proportionate to height	-do-
Constable	18 to 23 years	Matriculation/10 th / 12 th class pass		Relaxable for SC/ST/OBC in accordance with the instructions	Height (cms) Male -170 Female – 157	Relaxable by 5 Cms for candidates from Hill areas
					Chest - 80-85 Cms	77-82Cms for STs

				issued by the Govt from time to time.	Weight - Proportionate to height	-do-
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29. The aspirants seeking employment into the Police & CAPF must go through their respective websites of Police & CAPF to ascertain the latest enrolment conditions which are notified from time to time.

CONCLUSION

30. A candidate if desirous to join, mandatorily has to register individual online after entering his criteria. As and when the vacancies are allotted by UPSC, Central Staff Selection Commission, State Public Service commission or any other Government agencies, intimation to individual is given via Email or in website.

SUMMARY

31. The Indian Army has seven Commands ie. Northern, Western, Central, Southern, South Western, Eastern and Training Command.

32. The Navy is organised into three Commands i.e Western Naval Command, Eastern Naval Command and Southern Naval Command.

33. The Air Force is organized into seven commands i.e Western Air Command, Central Air Command, Eastern Air Command, South Western Air Command, Southern Air Command. Training Command and Maintenance Command.

34. The Combat Commands of Indian Army comprises Corps, Division and Brigades. These are commanded by an Officer of the rank of Lieutenant General, Major General and Brigadier respectively.

35. There are three major categories of army based on their role:-

- (a) Fighting arms - Armour, Infantry and Mechanised Infantry.
- (b) Supporting Arms- Artillery, Engineers, Army Aviation, Army Air Defence and Signals.
- (c) Supporting Services – Army Service Corps, Army Medical Corps, Army Ordnance Corps, Corps of Electronic and Mechanical Engineers etc.

36. Field Marshal is an honorary rank given to a General for his invaluable services rendered to our country.

37. Admiral of the Fleet is an honorary rank given to an admiral for his invaluable services and will continue to serve the rest of his term with the honorary rank.

38. Marshal of the Air Force is an honorary rank given to an Air Chief Marshal for his invaluable services.

39. Indian Armed Forces render honours and awards to the personnel excelling during peace and war.

40. Entry into Police Services as an Officer.

- (a) Passing All India Civil Services Examination conducted by All India UPSC Civil Services Examinations.
- (b) Promotions from State Cadre Police Services.

41. The other Police organizations Central Investigation and Intelligence Agencies are also function.

42. **Central Armed Police Forces Cadre and Personnel.**

- (a) Gazetted Officers.
- (b) Subordinate Officers.
- (c) Constables.

CHAPTER I: INTRODUCTION TO MAP READING

MAPS, CONVENTIONAL SIGNS, SCALE AND GRID SYSTEMS

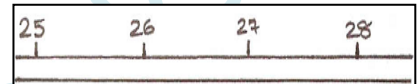
Definition of Map

1 A map represents selected natural and manmade features of the whole or part of the earth's surface on a sheet of paper. It has a definite scale and correct relative geographical positions and elevations. Symbols, colour differences and contours on map help to show the physical features *i.e.* mountains, valleys and plains. Maps show important natural and cultural features such as relief, vegetation, water bodies, cultivated land, settlements, and transportation networks, etc. These maps are prepared and published by the National Mapping Organisation of each country. The science of making maps is called as Cartography. For example, the Survey of India prepares the topographical maps in India for the entire country. A map, however, has one major limitation that it cannot show everything that exists on the ground.

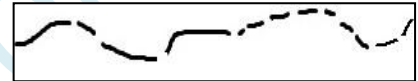
Conventional Signs

2 Conventional signs are symbols used to represent certain artificial or natural features/objects on the map. Some common types of conventional signs are listed as follows in Figure - 1

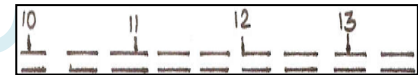
(a) Roads-metalled with Km-stone.



(b) Roads-unmetalled with Km-stone.



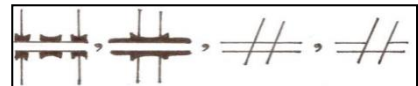
(c) Cart track, camel track, mule path.



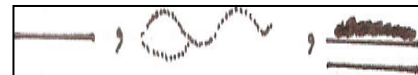
(d) Footpath, road in bed of stream, Level crossing.



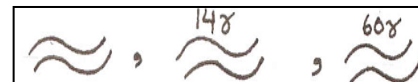
(e) Bridges with pier sand without, Causeway, Ford.



(f) Stream-Approx water course, canal River banks, shelving, steep 10 to 20 feet,, over 20 feet.



(g) River beds-dry, with stream, With island and rocks.



(h) Tidal river-shoal-submerged rocks.



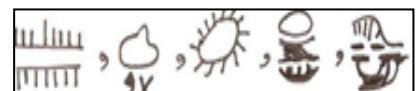
(i) Wells-lined and unlined, spring, Tanks- perennial and dry.



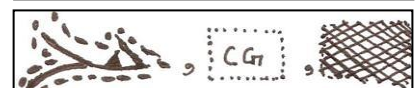
(j) Kaeaz - in, flow and dry, swamp, Reeds.



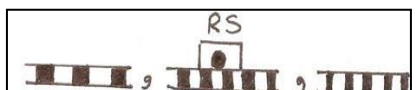
(k) Embankments, road or rail, tank cutting tunnel.



(l) Broken ground, camping ground, Vineontrellis.



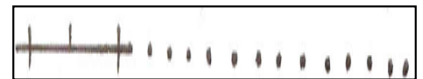
(m) Railways, broad gauge, Double, Single (Station), under construction.



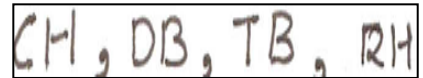
(n) Railways other Gauges, double, Single (milestone) and under Construction.



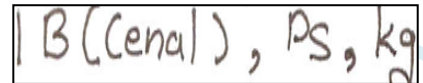
(o) Light railway or tramway, Telegraphline.



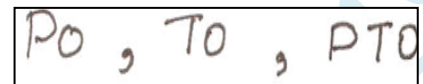
(p) Circuit house, Dak, Travellers, Bungalow, Rest House.



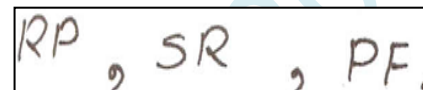
(q) Inspection bungalow, Police station, Buddhist Kyaung.



(r) Post office, telegraph office, Combined office.



(s) Forest-reserved, state and protected.



(t) Spaced names, Administrative, Locality, tribal.



(u) Villages: open, walled, ruined, Deserted antiquities.



(v) Huts, permanent and temporary, Fort, Tower chhatvi.



(w) Church, Mosque, Temple, Pagoda, Idgah, tomb.



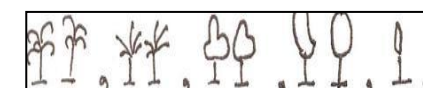
(x) Dams, masonry and Earthwork-work.



(y) Grass high and low cane, Bamboo plantation.



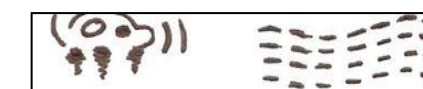
(z) Palms, Areca, palmyra, Other conifer, other trees, scrub.



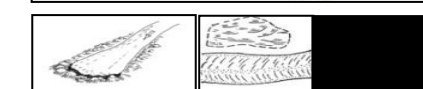
(aa) Contours, Form lines, Rocky slopes.



(ab) Cliffs-sand features.



(ac) Moraine, Glacier, Screen.



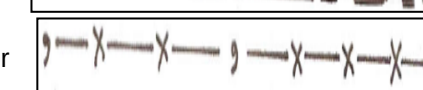
(ad) Boundary demarcated; International.



(ae) Boundary demarcated; Province or state.



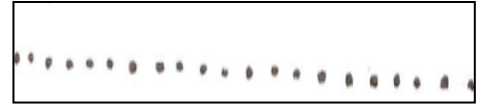
(af) Boundary un-demarcated; International, province or state.



(ag) Boundary; district or Tribal.



(ah) Boundary; Sub Divisional, Tehsil, Taluka or township forest.



(aj) Boundary pillars, surveyed, not found.



(ak) Graves, oil wells, Mine-Battle field with year.



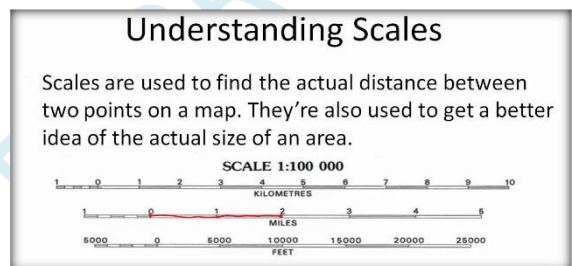
(al) Heights triangulated, Bench Mark.



SCALES AND GRID SYSTEMS

Definition

3 Scale is the proportion which the distances between the two points on the map that relates to the distance between two points on the ground. Everything on the map must be reduced and the extent to which the size is reduced makes the scale of the map.



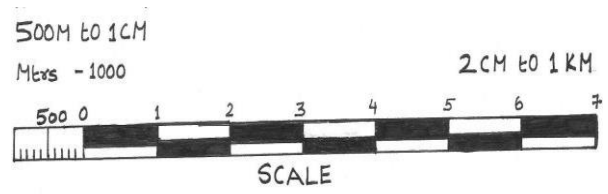
Methods of Expressing a Scale

4 There are two methods of expressing a scale:-

- (a) **In Words.** 1 inch to 1 mile, it means that 1 inch on the map represents 1 mile on the ground.
- (b) **As a Representative Fraction (RF).** This is the scale expressed in the form of a fraction. If the scale of a map is given as $1/100000$ this means that one unit of the map represents 100000 of the same unit on the ground. It could mean that one centimetre on the map represents 100000 cm on the ground.

Scale Line

5 Below the scale is the scale line by means of which distance on the map can be measured. In this scale 2 cm on map is equal to 1 km on ground. An example of the scale line for a scale "2 cm to 1 km" is as shown.



Definition of Grid and Grid Lines

6 The 'Grid' is a systematic pattern on Earth by laying a vertical and horizontal grid over the Earth's layout. The vertical lines are called the longitude and the horizontal lines are known as the latitude. Combinations of these lines are known as Grid Lines.

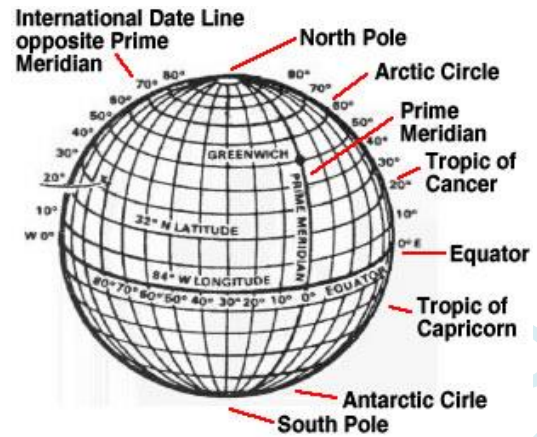
Purpose

7 The purpose of Grid Lines is to make possible giving and reading Grid References and to facilitate measurement of bearings.

Method of Grid Reference

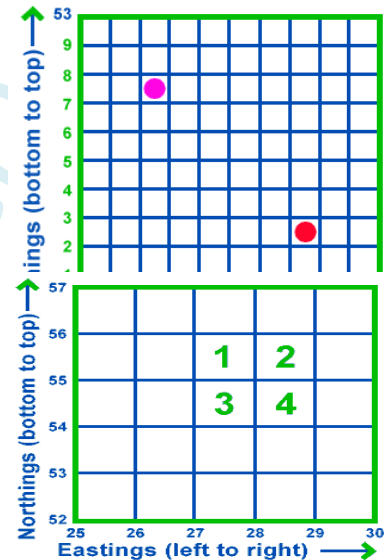
8 In giving a Grid Reference following rules should be remembered:-

- (a) A reference must always contain an even number of figures, normally it contains six figures.
- (b) EASTING lines are the black colour vertical lines.
- (c) NORTHING lines are the black colour horizontal lines.
- (d) Always count along the **EASTING** lines first from the WEST to EAST and then **NORTHING** from SOUTH to NORTH.
- (e) Grid References are of different types viz. Four Figure, Six Figure, Eight Figure and Ten Figure.
- (f) Mostly Six Figure Grid Reference is used.
- (g) For six figure Grid Reference the third and the Sixth figure represent the divisions of 1000 meters square to the nearest 10th part, so they have to be estimated and for these figures a slight latitude is allowed.
- (h) If a general Grid Reference is to be given or there is only one such object in one square e.g. bridge, temple, road junction then its identity and four figure grid reference would suffice.



9. Example

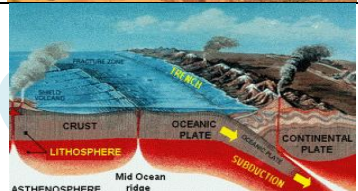
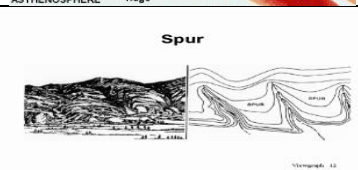
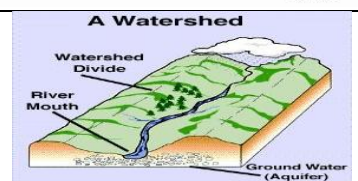
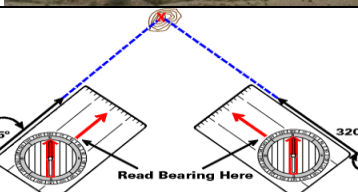
- (a) As we already said when giving a four figured grid reference, always give the Easting number first and the northings number second. In the diagram, the number 4 is in square 28 across (on the horizontal) and square 54 up (on the vertical) and therefore, the four-figure grid reference is '2854'.
- (b) The other number in the square above would get the following grid:- 2755; 2855; 2754; 2854.
- (c) In order to be little more precise with your grid references, you can give a 6 figure grid reference as shown in diagram 5b.
- (d) Here, we have taken the lower right square from the previous diagram and divided it by 10 in each direction. The circle is in the four-figure grid reference square '2552', but more accurately it is 2 tenth across and 7 tenths up with in that enlarged grid square, therefore the six-figure map reference is '252527'. The circle has 6 figure grid references of 257522.

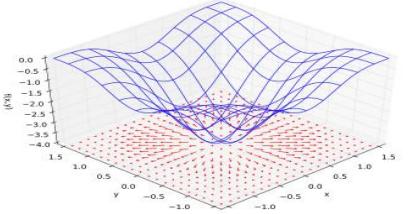
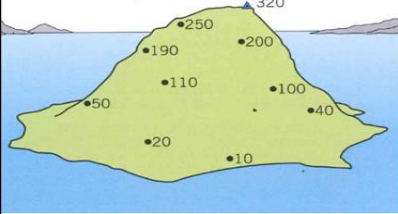
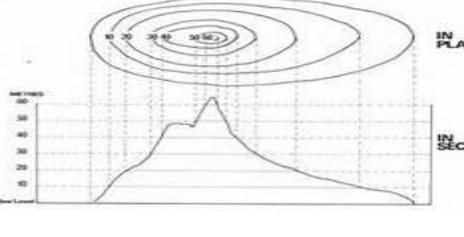
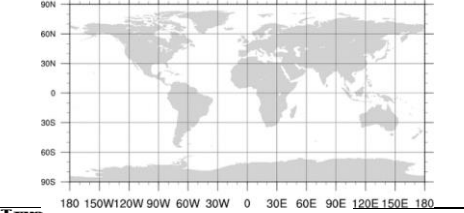
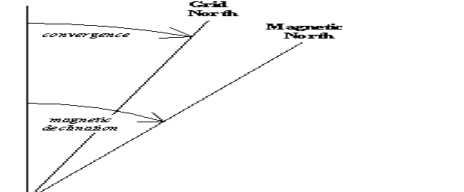
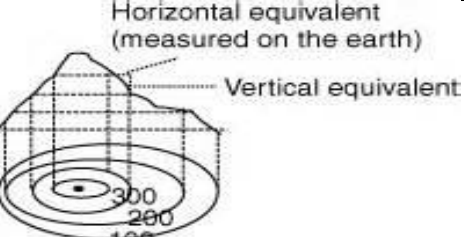


Topographical Forms and Technical Terms

10. Topographical forms are names used to describe geographical features which occur on the ground. The following are more commonly used:-

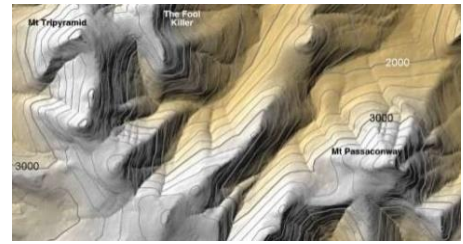
S No	Topographical Forms	Samples
(a)	Basin: An area of fairly level ground surrounded by hills or the area drained by a river or its distributaries.	
(b)	Col or saddle: A narrow ridge of high land joining up to higher hills.	

(c)	Crest: A highest part of hill or mountain range. It is that line on the range of hills or mountains from which the ground slopes down in opposite direction.	
(d)	Dead Ground: Ground which because of undulations or hills is not visible to the observer.	
(e)	Knoll: A small isolated hill.	
(f)	Plateau: A table land, an elevated region of considerable extent generally of same level.	
(g)	Ridge: A line along a hill or range of hills or mountains from which water flows in opposite directions.	
(h)	Spur: A piece of high ground jutting out of range of hills into lower ground.	
(i)	Watershed: The line separating the water flowing in two different rivers systems, the edge of a river basin.	
(j)	Defile: Any feature whether natural or artificial which could cause a body of troops to contract its front. An example of a natural defile is mountain pass while bridge is an example of an artificial defile.	
(k)	Escarpment: The steep hill side formed by a sudden drop in the general ground level usually from a plateau.	
(l)	Bearing: The angle formed by a line joining two points and the North and South line. Bearings are always measured clockwise.	

(m)	Bench Mark: A permanent mark usually cut into a wall recording exact height for future reference. It is marked as BM with height on Ordnance Survey Maps.	
(n)	Gradient: The slope of a hill expressed as a fraction.	
(o)	Spot Height: A point on a map whose height has been determined by Survey methods. These are usually shown as block dot with a number giving exact height above sea level in meters.	
(p)	Trigonometric Point: A point fixed during the triangulation at the beginning of a survey, marked on Ordnance Survey Maps by a small triangle with the height.	
(q)	Contours: A line drawn on the map joining up all points of equal height above sea level.	
(r)	Grid Lines: Lines running parallel to and at right angle to a North and south or East and West, Grid North is the direction of the North South grid lines on a map.	
(s)	Magnetic Variation: The difference between True North and Magnetic North.	
(t)	Horizontal Equivalent: The distance measured on the map between adjacent contour lines. It varies according to the nature of the relief.	

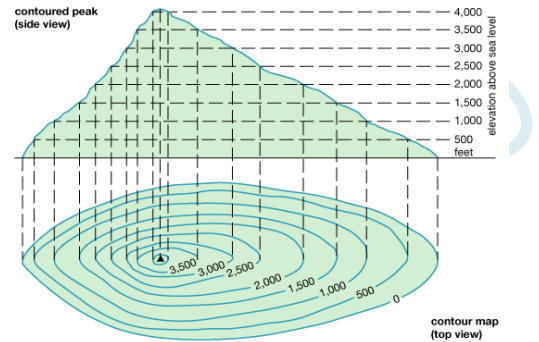
Relief, Contours and Gradients

11. **Relief** means the shape of the ground in a vertical plane. Representation of a relief on a map means showing of heights and shape of the ground above or below or datum which is normally sea level. **Relief** is shown with means of hachure, shading, form lines, layer tints, contours, spot heights, trig heights, bench marks and relative heights.

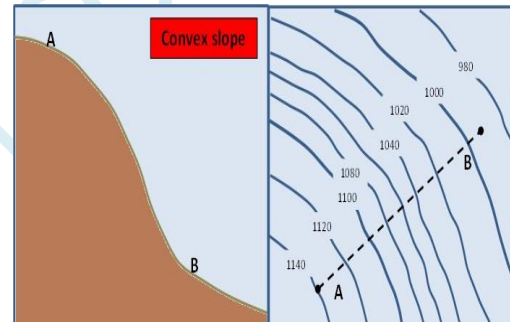
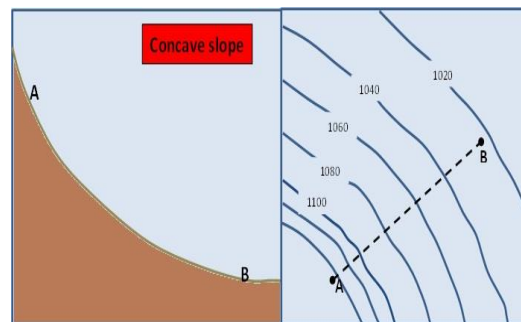


12. **Contour** is an imaginary line drawn on map showing same height from mean sea level. If you walk along a contour line you neither gain nor lose elevation. The Contours have following characteristics:-

- (a) Contours accurately show height, shape and slope of the ground.
- (b) Contours are shown generally in brown.
- (c) Height is marked on every fifth contour.
- (d) Contour lines vary in appearance.
- (e) These lines never touch or cross each other.



13. **Slope** means a surface of which one end or side is at a higher level than another. Contour lines represent to slope that is, closer the contour lines are, the steeper is the slope of the hill. If representation of slopes the contour lines are far apart, the slope down is gradual. The slopes are of two types, convex and concave. A convex slope is the one which bulges outwards and concave slope is the one which curves inwards.



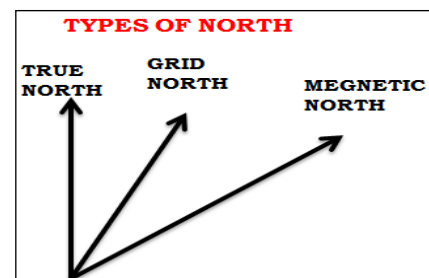
14. **Gradient** is the slope of the ground expressed as the angle the ground makes with the horizontal. A gradient of 1 in 15 means that in a horizontal. The horizontal equivalent is obtained by measuring on the map and vertical interval by subtracting the contour heights. You may often need to know just how steep a piece of ground is, whether a road is too steep for a certain type of vehicle to negotiate. The gradient can be worked out quickly from a contoured map. The rise or fall of a slope can be expressed in following two ways:-

- (a) In an Angle or Degree of Slope.
- (b) The tangent of the Angle or Gradient.

Types of North

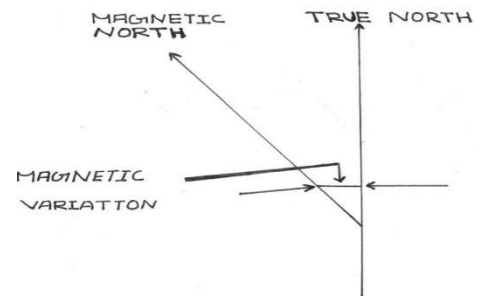
15. **Types of North** There are three types of North :-

- (a) **True North.** The direction of North Pole from the observer.
- (b) **Grid North.** North as per the Grid on map.
- (c) **Magnetic North.** It is the point to which a magnetic needle points, when freely suspended.



Magnetic Variation and Grid Convergence

16. **True North is Constant.** Magnetic North is the point to which the compass needle points. The needle does not point directly to True North, but a little West or East of True North. The point towards which the needle swings is known as Magnetic North and the difference between True North and Magnetic North is called Magnetic Variation. The amount of the Magnetic Variation depends upon two factors, time and place as at Fig below.



Magnetic Variation

17. **Time.** The Variation is not constant but is, gradually changing and even the change each year is not constant but the difference being negligible it is taken to be constant. On the top margin of a map will be found a statement giving the Magnetic Variation. To bring this up-to-date, the year of issue of the map must be noted and for every year that has passed since then the applicable change annually subtracted or added from the figure given as applicable.

18. **Place.** The amount of the Magnetic Variation also changes in different parts of the world and indeed in different parts of the country.

Grid Convergence

19. The angular difference between Grid and True North is called the Angle of Convergence or the Grid Convergence.

CONCLUSION

20. Knowledge about Cardinal points and types of North is the first step towards learning map reading. This knowledge is necessary not only with respect to maintaining direction during navigation; but also comes handy in our other daily activities. A good navigator has the ability to quickly orient himself as per the cardinal directions almost naturally; this helps in getting a sense of direction and helps to find out our own position subsequently.

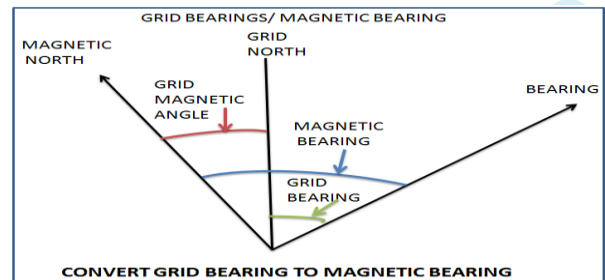
CHAPTER II : CONDUCT OF MAP READING, GOOGLE, TOURIST MAP AND APPLICATIONS

Introduction

1. Map reading plays very major role in services as well as in governance system. These days digital maps and artificial intelligence has taken over majority usage thereby making such task easy and user friendly along with information technology. Setting of map and finding own position is the essence of the map reading training.

Bearing and Its Conversion Methods

2. **Types of Bearing.** The clock wise angle formed by a straight line joining two points and direction of NORTH, is called the bearing between the two points. A bearing is always measured clockwise. They are three types as given below:-



(a) **Grid Bearing.** Measured on the map from the Grid North by the help of a protractor.

(b) **Magnetic Bearing.** Measured from Magnetic North by the compass.

(c) **True Bearing.** Calculated by finding out the relation of True NORTH and Grid NORTH or Magnetic NORTH.

3. **To Convert a Magnetic Bearing to a Grid Bearing.** Suppose the bearing of a certain point P is measured with a compass and is found to be 160° . To convert this Magnetic Bearing to a True Bearing, follow under mentioned steps:-

(a) First find out the Magnetic Variation of the Area. Magnetic Variation is given on the Top Right corner of each Map.

(b) Suppose 5° is the Magnetic Variation of the area. Now subtract this Magnetic Variation to the Magnetic Bearing.

(c) The resultant is the Grid Bearing i.e. 155°

4. **To Convert Grid Bearing to Magnetic Bearing.**

(a) Measure the Grid Bearing of an object on the map with help of the service protractor from your own position.

(b) Suppose the Grid Bearing of the object is 150° .

(c) Now, find out the Magnetic Variation of the area with the help of Map (Magnetic Variation is given on the Top right corner of the map). Suppose Magnetic Variation of the area is 6° .

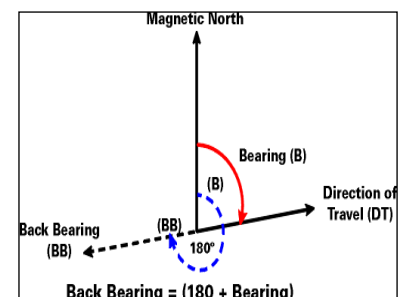
(d) Now, add this Magnetic Variation to the Grid Bearing.

(e) The resultant will be the Magnetic Bearing of the object i.e. 156° .

5. **Back Bearing.** It is bearing taken opposite of original position of object. The rule is that if the bearing is less than 180° add 180° and if bearing is more than 180° then subtract 180° . For example:-

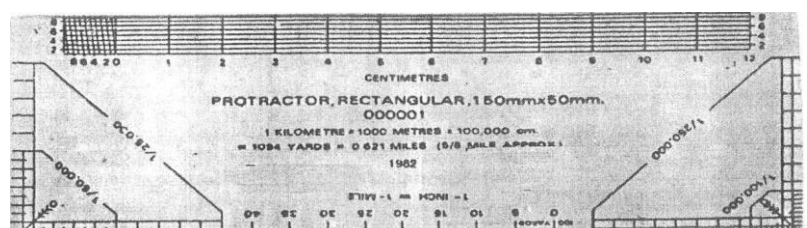
(a) If forward bearing of an object is 70° then its back bearing will be $180^{\circ} + 70^{\circ} = 250^{\circ}$.

(b) If forward bearing of an object is 240° then its back bearing will be $240^{\circ} - 180^{\circ} = 60^{\circ}$.



Service Protractor and its Uses

6. The protractor is made of cardboard or ivories (flexible material) and it measures 6 inches long and 2 inches wide.



Measuring a Bearing

7. The angle can be measured by drawing a line from the graduation to the point zero on the protractor. The required angle will be the gap between this line and the line joining the zero.

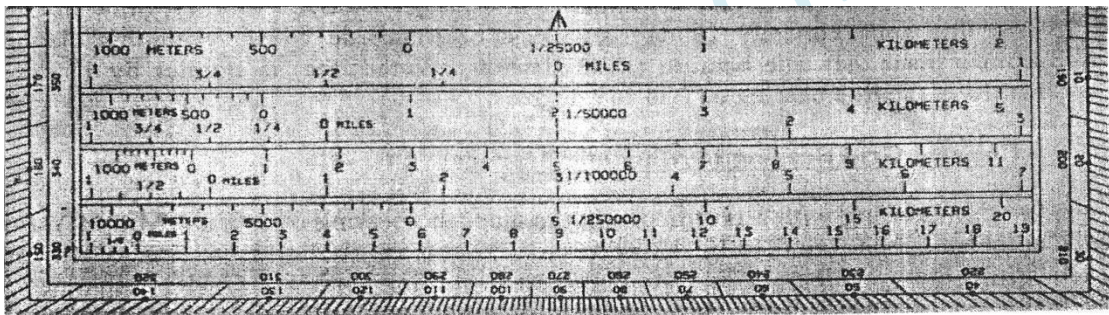
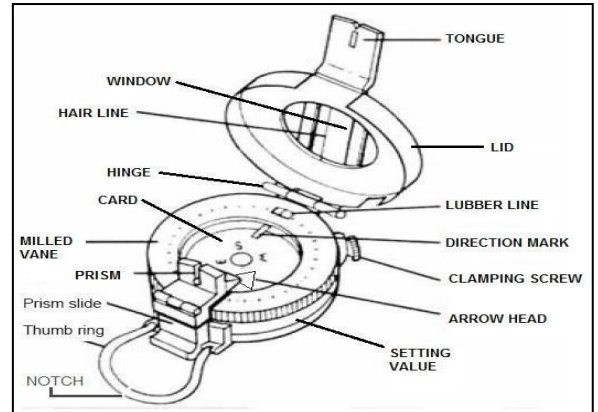
Types of Compass and Acquiring A Bearing including GPS

8. **Types of Compass and Taking Bearing.** There are two types of prismatic compass, the dry and liquid filled. Liquid type is easier to use though it is less sensitive.

9. **Description.** Various parts are shown below:-

10. **How to Take a Bearing**

- (a) Open the lid of compass.
- (b) Turn the prism casing over.
- (c) Put your thumb through the ring.
- (d) Put your forefinger underneath the compass & hold it to horizontal level.
- (e) Bring the prism up to the eye.
- (f) See through the prism via hairline to object.
- (g) Read the bearing.



Navigation by compass and GPS

11. **Navigation by Compass.** Sometimes it may be necessary to march by night or in thick jungle area in the direction of a certain point; since the point may not be visible in darkness or thick vegetation. In such a situation follow the steps mentioned below:-

- (a) Calculate the bearing from the Map i.e Grid Bearing
- (b) Convert it to the magnetic bearing and cater for the compass error to arrive at a figure (say 250 degree), the compass must now be set to this figure to march on it.
- (c) Unscrew the clamping screw
- (d) Rotate the milled vane to 25 till it comes exactly on the "lubber line"
- (e) Tighten the clamping screw
- (f) Compass is now set for 250 degree
- (g) To obtain the direction of march, open the lid fully and keep the compass on the palm.
- (h) Now turn left or right till arrow head comes under the direction mark
- (i) Direction of march is given by the direction in which tongue is pointing.

12. **Compass Error.** Sometimes due to the presence of impurities in the material of which a compass is made or other reasons, the magnetic needle may not point toward the magnetic NORTH but a little to the EAST or WEST of it. This deviation of the magnetic needle in the compass from the magnetic NORTH is termed compass error.

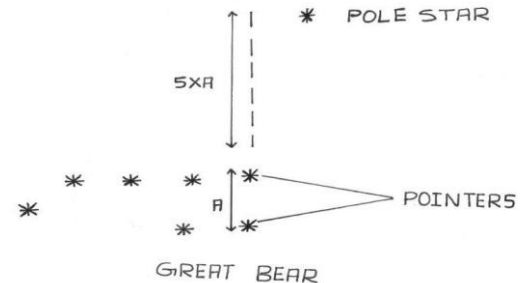
13. **Global Positioning System.** Global Positioning System (GPS) refers to a system of satellites and receivers that allows people and devices to pin point their precise location on the earth. The first GPS satellite was launched in 1974. GPS is funded and controlled by the United States, Department of Defence. Present technology provides very handy and accurate navigation. GPS is used by:-

- (a) Commonly used in day to day life by general public like.
- (b) Commonly used in day to day life by general public like for travelling purposes.
- (c) Fishermen and hikers to navigate.
- (d) Armed Forces, inbuilt its equipment and in uses in battlefields.

SETTING OF A MAP, FINDING NORTH AND OWN POSITION

Map Setting by Various Methods

14. **Setting of Map.** A map is said to be set or oriented when it is placed such that it corresponds directly with the ground i.e. when true NORTH on the map points to true NORTH on the ground. There are two methods of setting a map - by compass and by objects on the ground.



Setting by Compass

15. Draw a line showing magnetic NORTH from a point on a grid line. Open the compass and lay it flat on the map over the above drawn diagram, which will show the magnetic variation so that the hair line on the window lies along the magnetic NORTH line on the diagram. Then turn both the map and the compass till the needle points along the hair line. The map is now set, since the magnetic NORTH line on the map is pointing in the direction of magnetic NORTH as indicated by the compass needle.

16. **Finding North and own Position Without Compass.** The position of NORTH can be discovered by one of the following methods :-

(a) **Equal Altitude Method.**

- (i) Take a fairly large piece of paper or card board and spread it flat on the ground. In the centre fix a pencil or piece of wood perpendicular to the ground. It can be done with the help of a coin fixed at the base of pencil or wood with sealing wax or by directly pushing it in the ground.

- (ii) Wait till after mid-day until the sun has moved around sufficiently to throw another shadow as indicated by the dotted line AD i.e. of the same.

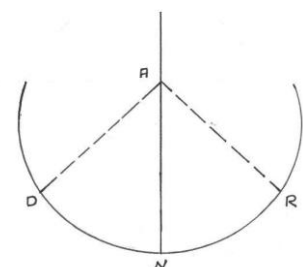
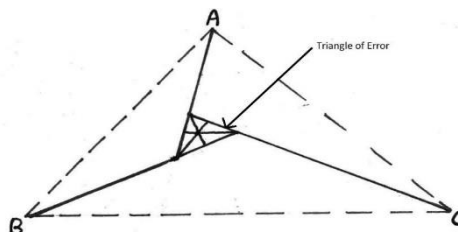
- (b) **By Stars.** In the Northern hemisphere, the Pole star indicates the position of True North to within 2 degree. It is a bright star and it can be found by protruding a line from Great Bear. The pole star will be found slightly off this line on the side remote from the remaining stars of the Great Bear.

17. **Methods of Finding own Position on Map.**

- (a) By resection method or Compass method.
(b) By Inspection method.

18. **Resection with Compass Method.**

- (a) Recognise three prominent features (A, B, C) on map and on the ground as well. These three prominent features must not be more than 180 or less than 30 apart. possible and clearly points be taken and bearings.



They should be as far as visible. The bearing of these converted into Grid

- (b) Then, on the map the back bearings from these points must be plotted, and the point of intersection will be the required position.

- (c) In order to do an accurate resection, three or more objects are necessary. But in that case if the three rays do not intersect at the same point, a triangle of error is obtained. The centre of triangle

is the point of your own position.

19. **By Inspection Method.** By inspections is meant a careful and detailed study of the ground and features both on the map and the ground and features on the map and on the ground. The method consists of:-

- (a) Setting the map.
- (b) Recognition of general area of own position on the map.
- (c) A close study of the ground details.

MAP TO GROUND, GROUND TO MAP

Map to Ground

20. To find out the details of map on ground is known as map to ground. Following methods are used to identify objects from map to ground:-

(a) **Bearing and Distance Method.** With the help of bearing and distance, find out own position. Find out the distance of the object to be identified on ground with the help of a scale on the map. Using service protractor, find out the bearing of the object and convert it into magnetic bearing. Set the magnetic bearing on compass and look for the object in the given bearing. Estimating the distance on ground the object will be identified.

(b) **Direction and Distance Method.** Draw a line on the map between own position and object to be identified. Calculate its distance and using any of the following methods find the direction of the object:-

- (i) With the help of a sight rule find the ground direction of the object.
- (ii) With the help of two points on the map estimate the ground direction.
- (iii) Place a foot ruler /pencil at own position and align it with line of the map.
- (iv) Place a pin each at own position and at the object on the map. Align both pins and find general direction.

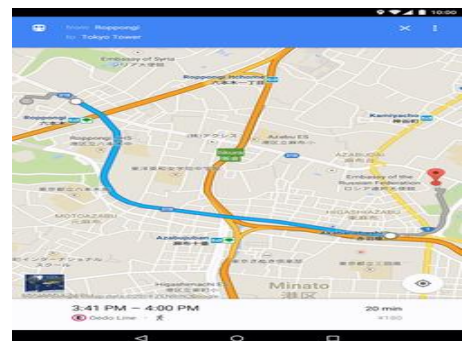
(c) **By Estimation Method.** In this method measuring bearing, distance and direction, object is identified with the help of other details in the proximity of the object.

Ground to Map

21. To find out an object indicated on ground on the map is called ground to map.

GOOGLE MAPS & APPLICATIONS

22. Google Maps is available as a mobile app for the Android and iOS mobile operating systems. The Android app was first released in September 2008, though the GPS-localization feature had been in testing on cellphones since 2007. Up until iOS 6, the built-in maps application on the iOS operating system was powered by Google Maps. However, with the announcement of iOS 6 in June 2012, Apple announced that they had created their own Apple Maps mapping service, which officially replaced Google Maps when iOS 6 was released on September 19, 2012. However, at launch, Apple Maps received significant criticism from users due to inaccuracies, errors and bugs.



Applications

23. Here are eight off-the-beaten-path Google apps that'll help you do all sorts of interesting things with your Android device. And yes, they're all free. (Some, but not all, are also available for iOS.)

- (a) Google Handwriting.

- (b) Trusted Contacts.
- (c) Google Trips.
- (d) Google Arts & Culture.
- (e) Chrome Remote Desktop.
- (f) Google Opinion Rewards.
- (g) Android Auto.
- (h) Wallpapers

CONCLUSION

24. It is important for the cadets to understand the methods and the procedure to find objects from map to ground and from ground to map. With the help of satellites and information technology companies like Google have taken over mapping system by using artificial intelligence making life faster and user friendly connecting Smart Phones with applications for map reading, navigation, tracking system and traffic.

SUMMARY

1. Map is the geographical representation of land on a paper.
2. Scale is a proportion of two points on map and two points on the ground.
3. Relief is applied to the shape of the ground in a vertical plane.
4. Contour is an imaginary line following surface of the ground at a certain level.
5. Gradient is the slope of the ground expressed as the angle the ground makes with the horizontal.
6. Cardinal Points are the four major directions North, South, East and West.
7. There are three types of North, True North is the direction of North pole. Magnetic North is the direction which a magnetic needle points, Grid North is the direction of the North South grid lines on a map-point.
8. Magnetic Variation is the difference between True North and Magnetic North. It depends on time and place.
9. Grid Convergence is the angular difference between Grid North and True North.
10. Bearing: The clock wise angle formed by a straight line joining two points and direction of North.
11. Methods to find own position on map are- by resection, by map spotting or by bearing and distance.
12. Methods are used to identify objects from map to ground.
 - (a) Bearing and Distance Method.
 - (b) Direction and Distance Method.
 - (c) By Estimation Method.
13. To find out an object indicated on ground on the map is called ground to map.
14. **Applications.**
 - (i) Google Handwriting.
 - (j) Trusted Contacts.
 - (k) Google Trips.
 - (l) Google Arts & Culture.
 - (m) Chrome Remote Desktop.
 - (n) Google Opinion Rewards.
 - (o) Android Auto.
 - (p) Wallpapers

CHAPTER- I : INTRODUCTION TO FIELD CRAFT CRAFT

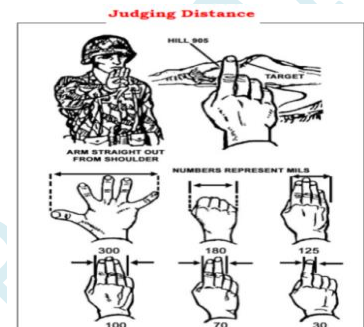
INTRODUCTION

1. Field Craft is an important aspect of military training as it is the art of using the ground and the weapon to the best of one's own advantage. A quick, accurate and standard procedure is necessary to enable a commander to describe an area to his men and the men to understand it correctly
2. Battle craft is nothing but set of drills which are essential for conduct of successful operations in the battle field. These battle drills are very useful in tackling minor tactical problems. They save time, ensure rapid action and avoid confusion. Knowledge of field signals and section and platoon formations, however, is essential in the execution of various battle drills.

Judging Distance

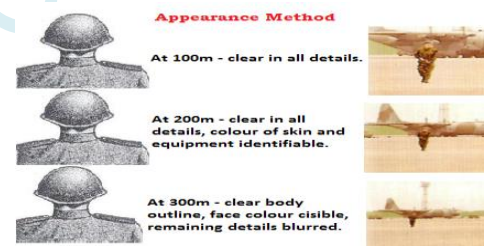
3. An individual should be able to judge distance accurately with his eyes so that the individual can decide on the following:-

- (a) Know when to open fire.
- (b) Know which weapon to be used.
- (c) Can indicate targets to other men in his section.
- (d) Pass back accurate information when acting as an observer.

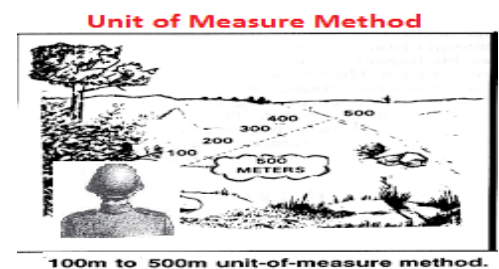


4. **Methods of Judging Distance.** There are six methods of Judging distance. These are as under:-

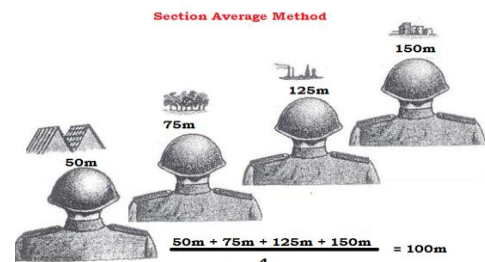
5. **Unit of Measure.** This method is also termed as the 100 yards method. The unit of measure chosen is normally 100 yards and therefore one should form a good idea of 100 yards distance on the ground. The length of a hockey field is the best yard stick for this purpose. The distance of a given object will be a multiple of the imaginary unit of 100 yards, as placed between the observer and the object.



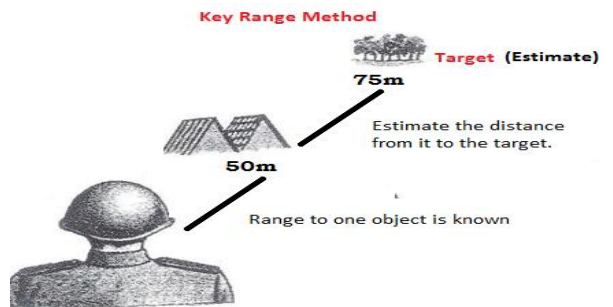
6. **Appearance Method.** The distance can be judged by noting the detailed appearance of man at various ranges.



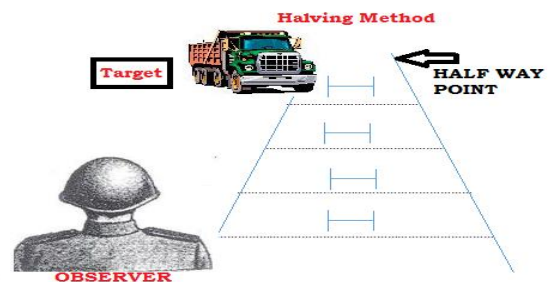
7. **Section Average.** Each man in the section is asked to judge the distance of a given object. The average of the answers given by the whole section is then accepted as the distance.



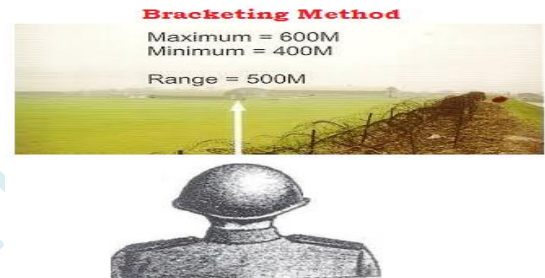
8. **Key Range.** If the range of the certain object is known, distance to other objects can be found in relation to the known range. This method is called "Key Range" method.



9. **Halving.** An object is selected half way between the observer and the target, the distance to the selected object is judged and doubled to get the distance to the target.



10. **Bracketing.** The observer works out the maximum and the minimum possible distances of the object and then accepts the mean as the distance.



CONCLUSION

1. To achieve success in war it is very important that the target is correctly understood for mechanics of field and battle crafts are understood clearly. Knowledge of ground along with basic skills would be beneficial to cadets in all respect.

CHAPTER II – INDICATION OF LANDMARKS

Introduction

1 Landmarks and other objects on the ground or a battle field may be either unclear due to climatic conditions or other reasons. Every effort should, therefore be made to indicate their location and size carefully and accurately

Definitions

2 **Landmarks.** An object, which is important on the ground and which is used in verbal orders to explain the ground in front.

3 **Target.** It is an object having a technical significance which is indicated with a view to bring down fire on it.

4 **Reference Point.** An important and unmistakable object, with the help of which you can indicate other land marks or targets. A reference point should be specific.

Types of Ground

5 **Broken Ground** It is uneven and is generally scattered with nullahs, bump sand fields in the ground. It is suitable for move of infantry.

6 **Flat and Open Ground.** It is even ground with little cover e.g. bushes, hedges and other foliage. It is not suitable for move of Infantry by day.

7 **High Ground.** Ground far above the general level of the area e.g. hill. It facilitates control of area around it by observation or fire.

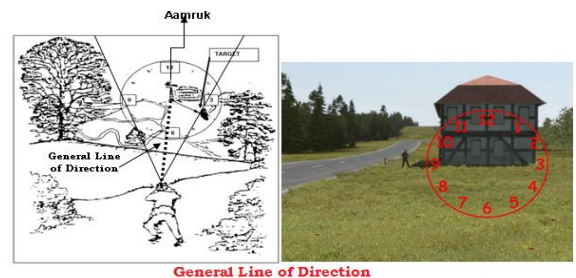
8 **Dead Ground.** Ground that is hidden from an observer's view. It cannot be covered by flat path weapons.

9 **Procedure Of Description.** The normal method of scanning and describing ground is by dividing it as follows:-

- (a) Fore Ground up to 300 yards.
- (b) Middle Distance from 300 yards to 500 yards.
- (c) Distance beyond 500 yards.

Indication of Landmarks

10 **General Line of Direction.** Start by giving the **general** line of direction by pointing out a centrally located, if possible, prominent land mark, e.g. No 1 section 500, RED HOUSE, class Red House and line beyond General Line of Direction (GLD).



11 **Boundaries.** After giving general line of direction give LEFT and RIGHT boundaries of your area. Divide the ground into foreground, middle and distance. Having done so start from LEFT to RIGHT systematically in a clockwise direction and describe the ground.

12 **Sequence of description.** While describing the ground bounded by particular arc after giving the boundaries start from LEFT to RIGHT. If the ground all around is to be described, start after general line of direction to the right and finish at general line of direction by completing the indication all around.

Methods of Indication of Targets

13 **Easy Targets.** Can be indicated by the following methods:-

- (a) **Indication by Description.** A noticeable target can often be described directly. For example "No. 1 Section BRIDGE". Here BRIDGE is so noticeable that nobody can make a mistake in recognizing it.

(b) **Indication by Direction or Range or Both.** In slightly less obvious cases other aids should be used e.g. direction or range or both. An example of each is given below:-

- (i) Indication by Direction :No 1 Section BAEN BAGHICHA.
- (ii) Indication by Range :No 1 Section 600 BAGHICHA.
- (iii) Direction and Range :When indicating a landmark, indicate direction first and then range e.g. BAEN-600, BAGHICHA.

14. **Difficult Targets** The targets which cannot be indicated by the methods given above are termed difficult targets. The methods to indicate these are explained by Direction Method as given below:-

- (a) The general line of direction.
- (b) A known reference point.
- (c) Another landmark.

CONCLUSION

15. To achieve success in war it is very important that the target is clearly understood and recognized by the troops. By using the methods of indicating the target the cadets can easily indicate and identify the target.

CHAPTER III: OBSERVATION, CAMOUFLAGE AND CONCEALMENT

Introduction

1. The term camouflage comes from French word “Camoufler” meaning “to blind or veil”. It means to disguise an object in plain sight in order to conceal it from something or someone. The object loses its three dimensional qualities and appears flat. Camouflage is defined as action of misleading enemy by concealing or misrepresenting the identity of own troops, equipment, installations and activities.

2. **Why Things are seen.** Various factors responsible for things to be seen are as follows:-

- (a) Shape.
- (b) Shine.
- (c) Shadow.
- (d) Surface.
- (e) Spacing.
- (f) Smoke.
- (g) Sound.
- (h) Movement.

3. You have seen as to why things are visible during day or in clear moonlit night. Now let us see why things are visible at night. There are two factors which are given below:-

- (a) Sound.
- (b) Light.

4. **Personal Camouflage.** Camouflage means to disguise an object in plain sight in order to conceal it from something or someone. Personal camouflage can be done by following methods:-

- (a) Use of Disruptive Pattern.
- (b) Clothing and Local Vegetation.
- (c) Camouflage of Face.
- (d) Camouflage of Equipment.

5. **Camouflaged of Equipments:-**

- (a) Helmet. Use of Hessian Cloth & Use of Camouflage Net.



- (b) **Camouflage of Back Packs.** Packs have square outline which is broken by tying thin ropes on the packs and branches of local vegetation are stuck in the ropes.



- (c) **Camouflage of Rifle.** Metal parts of rifles are phosphate coated and hence shine is reduced. Garnish of suitable colour should be wrapped over wooden parts / plastic parts to contrast with surroundings.



- (d) **Camouflage of LMG.** Wooden/plastic parts of the LMG should be covered by garnish. LMG trench is camouflaged by use of a net.



- (e) **Camouflage of Equipment.** Equipment like binocular, map case, Radio Set should also be camouflaged by breaking its outline and preventing shine.



Concealment

6. When the terrain does not provide natural cover and concealment, you must prepare your cover and use natural and man-made materials to camouflage/ conceal yourself, your equipment, and your position.

7. **Types of Cover and Correct Use.** There are of following types:-

- (a) Cover from View.
- (b) Cover from Fire.
- (c) Look Through or Around Cover.
- (d) Avoid Breaking a Straight Line.
- (e) Isolated Cover is Dangerous.

CONCLUSION

8. To conclude, the importance of camouflage and concealment can be realized from the following:-
- (a) In earlier days it was said "If it can be seen, it can be hit, if it can be hit it can be killed".
 - (b) But now in the modern warfare "If it can be seen it will be killed".
9. Therefore, camouflage needs greater emphasis and the art of camouflage and concealment reduces the different varieties of soldiers into two main categories viz, "The good and the dead." Concealment is an aid to tactical deception and misinforms the enemy about our intentions and strengths.

CHAPTER IV : USE OF GROUND AND MOVEMENT

1. Once we are forced to movement in open, a part of our force will have to fire on en position & force him to keep his head down. This would render the en incapable of bringing down aimed fire at us while we are on the movement. This process of keeping one element on the ground to give covering fire, while the other element is on movement, is called use of ground and movement or fire & movement.

2.

Field Signals

3. In Army, different methods are used to convey messages are given as under:-

SIGNALS WITH HAND AND WEAPONS

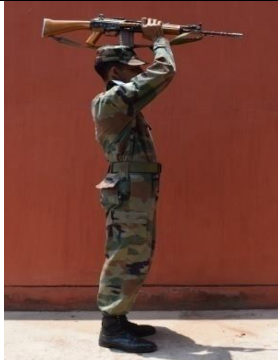
4. Signals with Hands.

Ser No	Name Of Signal	Signals With Hand	Illustrations
(a)	Deploy	Right arm fully extended above head and waved from side to side, palm open.	
(b)	Advance	Right arm swung from rear to front in "under arm blowing" fashion.	
(c)	Halt	Right arm raised to full extent above head.	
(d)	Turn About	Right arm raised and bent above head.	

(e)	Change Direction	Right arm raised to front in line with shoulder. Body then turned in required direction.	
(f)	Close	Right hand place on top of head, elbow to the right.	
(g)	Follow me	Right arm swung from rear to front above the shoulder in "over arm bowling" fashion.	
(h)	Enemy Approaching	Both hands open, palm inwards at waist level, with inwards scooping motion.	
(i)	Enemy LMG firing	Right hand thumb down signal.	

(j)	Attack	Punching motion with Right or Left hand according to direction of attack.	 
(k)	Closed to Rendezvous	Close sign followed by both hands clasped in front of body at waist level.	 
(l)	Infantry obstacle ahead	Both hands crossed in front of body at the waist, palm open downwards.	

5. **Signals with Weapons.**

Ser No	Name Of Signal	Signals with Weapons	Illustrations
(a)	Enemy in Sight	Rifle held above the head parallel to the small number muzzle in the direction of the enemy.	
(b)	Enemy in Sight in large number	As per (a) above, but arm moved up and several times.	 

(c)	Advance	Both arms raised to form the letter U	
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6. **Signals with Whistle.**

Ser No	Name Of Signal	Signals With Hand
(a)	Cautionary Blast	A short blast to draw attention to a signal or order about to be given.
(b)	The Alarm Blast	A succession of alternate long and short whistle blasts.
(c)	Enemy Aircraft	A succession of short blasts.
(d)	Enemy Aircraft departed	Two long blasts repeated at interval of five seconds.

Field Signals As Means Of Giving Orders

7. There are various occasions when voice control is not possible. They are:-

- (a) Battle Noises.
- (b) **Need for Silence.** There are certain operations which, by design are carried out in utmost silence, eg:-
 - (i) Ambush.
 - (ii) Patrolling.
 - (iii) Raid.
 - (iv) Cordon.
- (c) Intervening Distances are Too Large.

8. **Methods to Attract Attention Of Troops.** Before any field signal is executed, the commander has to attract the attention of troops. Methods to attract attention of troops are :-

- (a) A Short Blast of Whistle.
- (b) A Bird Call.
- (c) Whistle by Mouth.
- (d) Clicks (By using tongue).
- (e) Clicks by Fingers.

9. **Other Methods of Communication.** Besides field signals there are various other means of communication in the Army which are as follows:-

- (a) Dispatch Runners.
- (b) Dispatch Rider.
- (c) Radio Sets.

10. **Field Signals By Day.** Some of the visual signals used during the day are flags and mercury coated mirrors. Flags are very effectively used by Navy on board a ship by following ways:-

- (a) Flags: (Red, Green and White flags).
- (b) Mercury coated mirrors.
- (c) Smoke.
- (d) Miscellaneous: Various signals can be improvised and pre-arranged as under:-
 - (i) Clothes superficially hung out to dry.
 - (ii) Hurricane lamp, kept in the window.
 - (iii) Flashing of torch is used as Morse Code.
 - (iv) Applying various colours / signs on forehead and arms.

11. **Field Signals By Night.** Some of the field signals that can be used at night are:-

- (a) Pre decided signals on a walkie talkie.
- (b) Click by fingers.
- (c) Clicks by using tongue.
- (d) Whistle by the mouth.
- (e) Use of rope.
- (f) Use of colour light.
- (g) Use of blacked out torch.
- (h) Firing of weapon.

FIRE AND MOVEMENT

1. **When to use Fire and Movement Tactics.** Fire & movement tactics may be used in following circumstances: -

- (a) The enemy has opened SA fire which is effective.
- (b) When own troops have seen the en first - within 400 to 700 meters.
- (c) When the en is known or suspected to be in a certain area, then fire& movement tactics may be adopted when the troops reach within the effective range of en weapons/observation. (Instructor to explain as to what could be the effective range of enemy's personal weapon)
- (d) To cross obstacles by day or by night, e.g. nullahs /rivers.

2. **Selection of Fire Positions and Fire Control.** The ideal fire position should:-

- (a) Provide cover from fire.
- (b) Provide cover from view.
- (c) Afford a good view of the ground to be watched or target to be engaged.
- (d) Provide room in which to use the weapon freely.
- (e) Have a covered approach.
- (f) Be easy to advance from.

CONCLUSION

3. Fire & Move is the basic skill of tactics. Its training provided to all the officers and jawans. The basic aim is to enable one detachment of soldiers to move while the enemy is pinned down by the effective fire of the other temporarily static body of soldiers. It is the most effective and safest method to move during operations or when in contact with the enemy.

CHAPTER V : SECTION FORMATIONS

Introduction

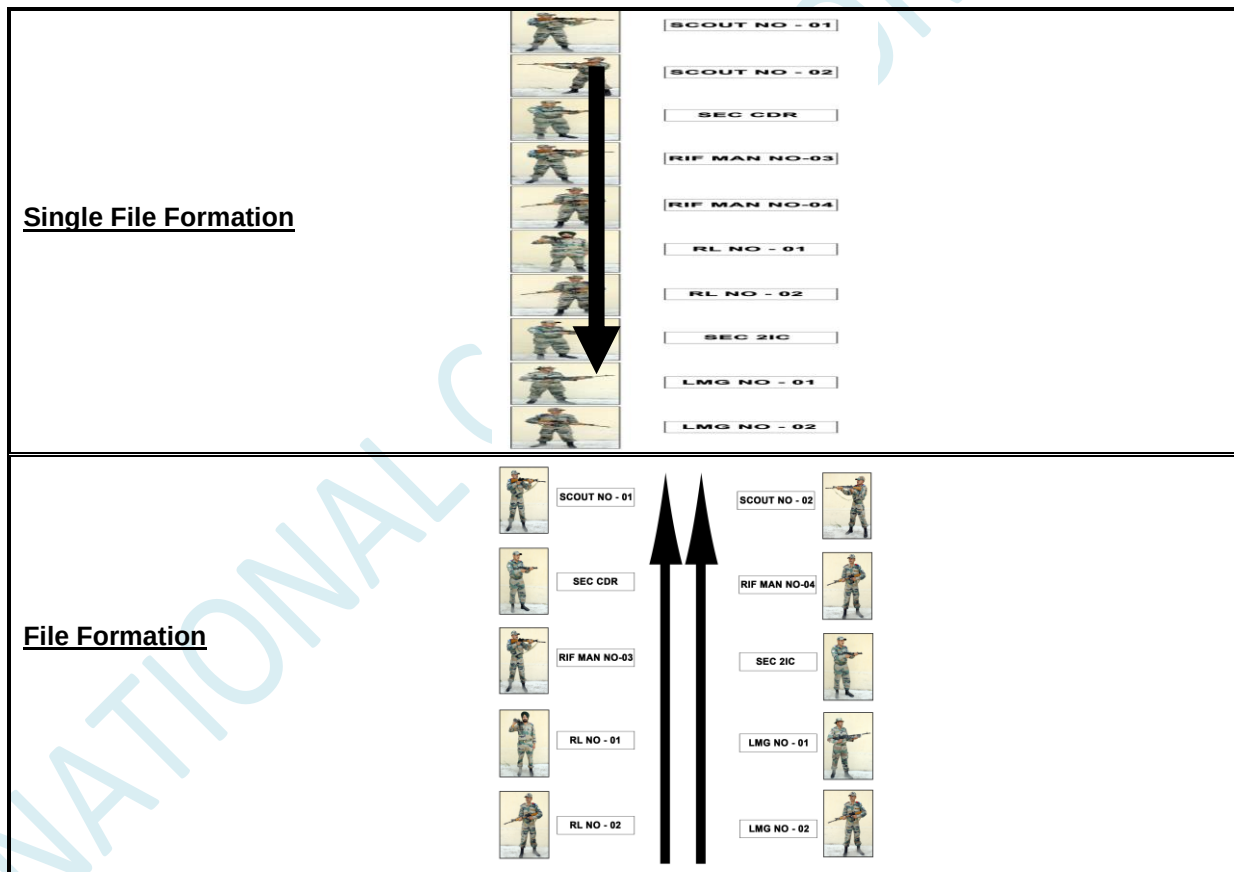
1 After learning about use of ground and movement, the cadet must know section formations adopted and used. Such formations are adopted keeping in mind the best use of ground for our own advantages as well as bringing effective fire on enemy if weapons are given.

Section Formation

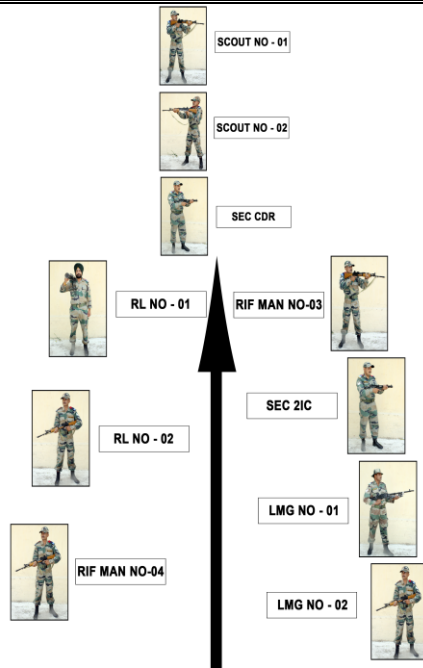
2 **Introduction.** Various formations are used when troops come in contact with the enemy and the type of formation adopted is entirely dependent on the following four basic factors:-

- (a) Degree of control required to be exercised by the Section Commander
- (b) Type of ground.
- (c) Necessity of bringing down maximum fire with minimum delay.
- (d) Task.

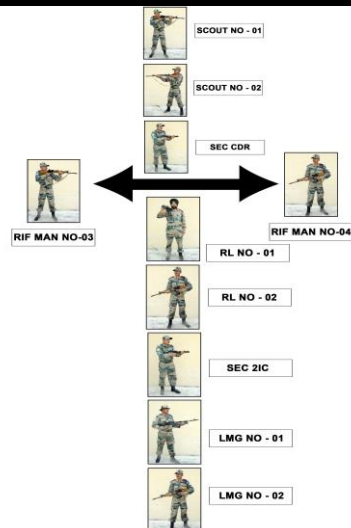
3 A Section is the smallest sub unit of an Infantry Battalion and is capable of undertaking independent task. It consists of ten persons who are organized in Rifle group and Support Group. Different types of formations adopted by a section are as under:



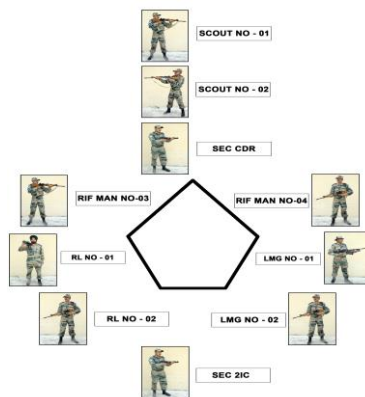
Arrow Head Formation

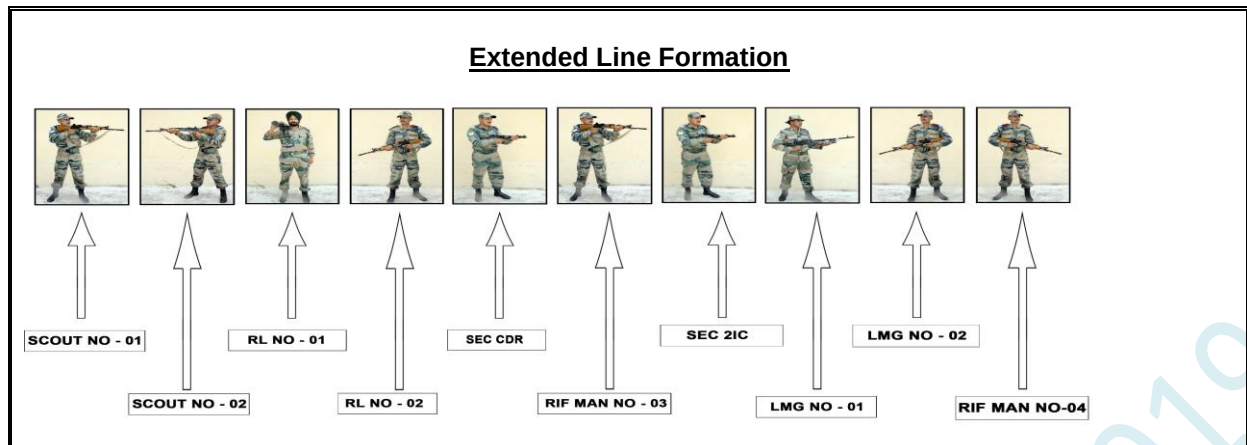


Spear Head Formation



Diamond Formation





Scout

4. Now you have seen section formation, a word about the scouts. Scouts are the eyes and ears of the section. Scouts always work in pairs. They work ahead of the leading section and advance from bound to bound. As scouts, one must always be alert.

FIRE CONTROL ORDERS TO A SECTION

Importance Of Fire Discipline And Fire Control Orders

5. Fire discipline is a battle winning factor which will stand you in good stead, especially in situations where surprise is of paramount importance. Indisciplined firing starts with an individual and spreads like wild fire. Should the troops have confidence of correct, accurate and effective fire orders, they may not indulge in opening of premature/indiscriminate firing, thereby giving away position or wasting ammunition. Fire discipline and fire control orders are very important in all operations of war.

6. **Important Terms.**

- (a) Fire Unit
- (b) Fire Direction Orders
- (c) Fire Control Orders
- (d) Arc of Fire

7. **Points for Section Commander.**

- (a) Indication
- (b) Range
- (c) Best Weapons to Use.
- (d) Rate of Fire.

CONCLUSION

4. Fire & Move is the basic skill of tactics. Its training provided to all the officers and jawans. The basic aim is to enable one detachment of soldiers to move while the enemy is pinned down by the effective fire of the other temporarily static body of soldiers. It is the most effective and safest method to move during operations or when in contact with the enemy.

CHAPTER V : KNOTS, LASHINGS AND STRETCHERS

Introduction

1. The ability to join two pieces of natural material together, and so increase their length, gives man the ability to make full use of many natural materials found locally. Knot tying is a useful exercise to obtain better coordination between eyes and fingers.

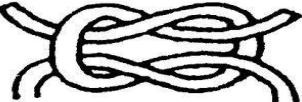
Knots

2. A brief description of the use to which the knot may be put is given in this lesson. The diagrams will explain how the knot is tied. The letter "F" means the free or untied end of the rope, and the letter "S" means the standing or secured end.

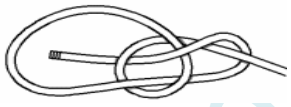
3. Knots for Rope ends or for Grips on Thin Rope.

S No	Type of Knot	Sample
(a)	Thumb Knot. To make a stop on a rope end, to prevent the end from fraying or to stop the rope slipping through a sheave, etc.	
(b)	Overhand Knot. Over hand knot may be put to the same use as the thumb knot. It makes a better grip knot, and is easy to undo.	
(c)	Figure Eight. This knot is used as the thumb knot. It is easy to undo, and more ornamental.	

4. Knots for Joining Ropes.

S No	Types of Knot	Sample
(a)	Reef Knot. To securely join two ropes of equal thickness together. Notice the difference in position of the free and standing ends between this and the thief knot.	
(b)	Thief Knot To tie two ropes of equal thickness together so that they will appear to be tied with a reef knot, and will be retied with a true reef knot. This knot was often used by sailors to tie their sea chests, hence the name.	
(c)	Fisherman's Knot. For joining two springy materials together; suitable for wire, fishing gut or vines. Two thumb knots (one on each rope) pulled tight. The knots lock together.	

5. **Knots to Make Loops in Rope.**

S No	Types of Knot	Sample
(a)	<u>Bowline.</u> To form a loop that will not slip on a rope end.	

6. **Knots for Fastening Ropes.**

S No.	Type of Knot	Sample
(a)	<u>Clove Hitch.</u> For securing a rope to as par. This hitch, if pulled taut, will not slip up or down on a smooth surface. A useful start for lashings.	

Lashings

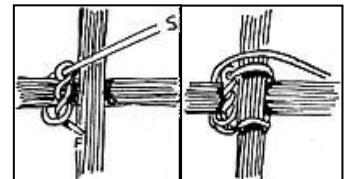
7. The methods employed to tie with ropes, poles or any rope to a stationary object to securely hold it in place is known as lashing.

(a) **Square Lashing.**

To join poles at right angles.

(b) **Frapping Turns.**

These are turns that go round the lashing and pull it tight.



8. **Stretchers.** A stretcher, litter, or pram is an apparatus used for moving patients who require medical care. A basic type (cot or litter) must be carried by two or more people. A wheeled stretcher (known as a gurney, trolley, bed or cart) is often equipped with variable height frames, wheels, tracks, or skids.

9. **Basic stretchers.**

(a) Simple stretchers are the most rudimentary type. They are lightweight and portable, made of canvas or other synthetic material suspended between two poles or tubular aluminum frame. Many are stored as disaster supplies and are often former military equipment.



(b) The scoop stretcher is used for lifting patients, for instance from the ground onto an ambulance stretcher or onto a spinal board. The two ends of the stretcher can be detached from each other, splitting the stretcher into two longitudinal halves. To load a patient, one or both ends of the stretcher are detached, the halves placed under the patient from either side and fastened back together.

(c) The litter, also known as a rescue basket or Stokes basket, is designed to be used where there are obstacles to movement or other hazards: for example, in confined spaces, on slopes, in wooded terrain. Typically, it is shaped to accommodate an adult in a face up position and it is

used in search and rescue operations. The person is strapped into the basket, making safe evacuation possible. The litter has raised sides and often includes a removable head/torso cover for patient protection. After the person is secured in the litter, the litter may be wheeled, carried by hand, mounted on an ATV, towed behind skis, snowmobile, or horse, lifted or lowered on high angle ropes, or hoisted by helicopter.

(d) A Reeves Sleeve, SKED, or "flexible stretcher" is a flexible stretcher that is often supported longitudinally by wooden or plastic planks. It is a kind of tarpaulin with handles. It is primarily used to move a patient through confined spaces, e.g., a narrow hallway, or to lift obese patients. Reeves stretchers have six handholds, allowing multiple rescuers to assist extrication.

(e) The WauK board is also designed for use in small spaces. The patient is secured to the board with straps. It has two wheels and a foldable footrest at one end, allowing the patient to be moved by one person, much as with a hand truck for moving cargo. It can also be used at a variety of angles, making it easier to traverse obstacles, such as tight stairwells.

10. **Wheeled Stretchers.** For ambulances, a collapsible wheeled stretcher, or gurney, is a type of stretcher on a variable-height wheeled frame. Normally, an integral lug on the stretcher locks into a sprung latch within the ambulance in order to prevent movement during transport, often referred to as antlers due to their shape. It is usually covered with a disposable sheet and cleaned after each patient in order to prevent the spread of infection. Its key value is to facilitate moving the patient and sheet onto a fixed bed or table on arrival at the emergency department. Both types may have straps to secure the patient.



11. **Other Types of Stretchers.** The Nimier stretcher (*brancard Nimier*) was a type of stretcher used by the French army during World War I. The casualty was placed on their back, but in a "seated position", (that is, the thighs were perpendicular to the abdomen). Thus, the stretcher was shorter and could turn in the trenches. This type of stretcher is rarely seen today.



CONCLUSION

12. Knots, lashings and stretchers are very useful and become very handy for cadets during camps. They can make use of knots for joining or tying 2 to 3 different ropes together and make use of it during rope climbing, rappelling, slithering and other such adventure activities. Similarly, lashings can be used for joining 2 or 3 things together to make a structure that can be useful in camps. For evacuation of any injured or casualty stretchers are important to be known and used.

SUMMARY

13. Field Craft is an aspect of military training which relates to the conduct of a soldier in face of the enemy. It's an art of using the ground and the weapon available to the best of one's own advantages.

14. **Methods of Judging Distance.**

- (a) Unit of Measure.
- (b) Appearance Method.
- (c) Section Average.
- (d) Key Range.
- (e) Halving.
- (f) Bracketing.

15. **Types of Ground.**

- (a) Broken Ground.
- (b) Flat and Open Ground.
- (c) High Ground.
- (d) Dead Ground.

16. **Landmarks.** .An object, which is prominent on the ground and which is used in verbal orders to explain the ground in front.

17. **Target.** It is an object having a tactical significance which is indicated with a view to bring down fire on it.

18. **Reference Point.** .A prominent and unmistakable object, with the help of which you can indicate other land marks or targets. A reference point should be specific. Things are seen due to

- (a) Shape.
- (b) Shine.
- (c) Shadow.
- (d) Spacing.
- (e) Smoke.
- (f) Sound.
- (g) Movement.

19. **Camouflage.** The various measures include.

- (h) Use of Disruptive Pattern Clothing and Local Vegetation.
- (i) Camouflage of Face.
- (j) Camouflage of Equipment.
- (k) Helmet.
- (l) Use of Hessian Cloth.
- (m) Use of Camouflage Net.
- (n) Camouflage of Packs.
- (o) Camouflage of LMG.
- (p) Camouflage of Rifle.

20. **There are two types of cover:-**

- (a) **Cover from Fire.** This implies that the concealed person is protected both from view and fire of weapon.
- (b) **Cover from View.** In this type of cover, a person is concealed only from view or from being seen not from fire.

21. Field Signals is one of the ways of passing of message with the help of Predetermined codes and signals.

- (a) Signals with hand.
- (b) Signals with Weapons.
- (c) Signals with Whistle.

22. Field signals are alternate means of giving orders and control troops when voice control is not possible like:-
- (a) Battle Noises.
 - (b) Need for silence.
 - (c) Intervening distances are too large.
23. The Field signals used during day are different from ones used at Night. These could be:-
- (a) Other Methods of Communication.
 - (b) Radio.
 - (c) Dispatch Rider.
 - (d) Runners.
24. The various section formations are :-
- (a) Single file.
 - (b) File.
 - (c) Arrow Head.
 - (d) Diamond.
 - (e) Spear Head.
 - (f) Extended Line
25. Knots are ability to join two pieces of material/rope together. Important types of knots are:-
- (a) Thumb Knot.
 - (b) Overhand Knot.
 - (c) Figure Eight Knot.
 - (d) Thief Knot.
26. **Lashing.** The method employed to tie with ropes, poles, or any rope to a stationary object to securely hold it in place is known as Square Lashing and Frapping Turns.
27. **Stretchers.** For evacuation of any injured or causality stretchers are important to be known and used.

CHAPTER I : BIOGRAPHIES OF RENOWNED GENERALS

Introduction

1. Military History is a humanities discipline, within the scope of general historical recording of armed conflicts in the history of humanity, and its impact on the societies, their cultures, economies and changing national and international relationships. The essential subjects of military history study are the causes of war, the social and cultural foundations, military doctrine on each side, the logistics, leadership, technology, strategy, and tactics used, and how these changed over time. Thus it is a dynamic discipline.

FIELD MARSHAL KODANDERA MADAPPA CARIAPPA, OBE

2. **Early Life and Military Career.** Cariappa was born at Shanivarsante in Kodagu (Coorg) which is currently in Karnataka. In 1919, he joined the first batch of KCIOs (King's Commissioned Indian Officers) at The Daly College at Indore and was commissioned into the Carnatic Infantry at Bombay as a Temporary Second Lieutenant. In 1927, Cariappa was promoted to Captain. He saw active service with the 37 (Prince of Wales) Dogra in Mesopotamia (present-day Iraq) and was later posted to the 2nd Queen Victoria's Own Rajput Light Infantry, which became his permanent regimental home. He was the first Indian officer to undergo the course at Staff College, Quetta in 1933. He was promoted to the rank of Major in 1938.

3. Cariappa served in Iraq, Syria and Iran from 1941–1942 and then in Burma in 1943–1944. He spent many of his soldiering years in Waziristan. He earned his 'Mentioned in Despatches' as DAA and QMG of General (later Field Marshal) Slim's 10th Division. He was the first Indian Officer to be given command of a unit in 1942. By 1944, Cariappa was a Temporary Lieutenant-Colonel. After command, he volunteered to serve in 26 Division engaged in clearing the Japanese from Burma, where he was decorated with an "Officer of the Order of the British Empire." In 1946, he was promoted as the Brigadier of the Frontier Brigade Group. Post-Independence, Cariappa was appointed as the Deputy Chief of the General Staff with the rank of Major General. On promotion as Lieutenant General, he became the Eastern Army Commander.

4. On outbreak of war with Pakistan in 1947, he was moved as General Officer Commanding-in-Chief, Western Command and directed operations for the recapture of Zojila, Drass and Kargil and re-established a linkup with Leh. In all this, he showed tremendous energy in moving troops, against considerable odds and finally ensuring success. On 15 January 1949 Cariappa was appointed as the first Indian Commander-in-Chief of the Indian Army. Cariappa was then instrumental in turning an Imperial Army into a National Army.

FIELD MARSHAL SAM MANEKSHAW, MC

5. **Early Life and Education.** Manekshaw was born on 3 April 1914 in Amritsar, Punjab to Parsi parents, Hormusji Manekshaw, a doctor, and his wife Hilla, who moved to Punjab from the city of Valsad on the coastal Gujarat. Sam's father served in the British Indian Army as a Captain in the medical services and also participated in World War I. Hormusji and Hilla had six children of which Sam was the fifth one. Fali, Cilla, Jan and Sehroo preceded Sam and Sam was followed Jemi, who later joined the air force as a doctor and was the first Indian to be awarded the air surgeon's wings from Pensacola, United States. After completing his schooling in Punjab and Sherwood College, Nainital, and achieving a distinction in the School Certificate of the Cambridge Board at the age of 15, he asked his father to send him to London to become a gynaecologist. But his father refused to send him to London stating that he was not old enough.

6. In an act of rebellion against his father's refusal, Manekshaw took the entrance examination for enrollment into the Indian Military Academy and was one of the fifteen cadets to be selected through open competition. He stood sixth in the order of merit. Just before taking over as the Chief of the Army Staff, at a function on 5 June 1969 to mark the centenary of Sherwood College, Manekshaw recalled that his years at the college had prepared him for war as they had taught him to live alone and independently, to fight without relent, tolerate hunger for long periods and to hate his enemy.

7. **Military Career.** Manekshaw's military career spanned four decades, from the British era and World War II, to the three wars against Pakistan and China after India's independence in 1947. He held several regimental, staff and command assignments. Manekshaw went on to become the 8th chief of the army staff, led the Indian Army successfully in a war with Pakistan and became India's first field marshal after independence. On commissioning, as per the practices of that time, Manekshaw was first attached to the 2nd Battalion, The Royal Scots, a British battalion, and was later posted to the 4th Battalion, 12th Frontier Force Regiment, commonly known as the 54th Sikhs. Manekshaw was later reassigned to the 16th Punjab Regiment, before being posted to the 3rd Battalion, 5th Gorkha Rifles, and then adopted 8th Gorkha Rifle after partition and went on to become the Colonel of the Regiment of 8th Gorkha Rifle.

8. **World War II.** During World War II, the then-Captain Manekshaw saw action in Burma in the 1942 campaign on the Sittang River with the 4th Battalion, 12 Frontier Force Regiment, and had the rare distinction of being honoured for his bravery on the battlefield. During the fighting around Pagoda Hill, a key position on the left of the Sittang bridgehead, he led his company in a counter-attack against the invading Japanese Army and despite suffering 50% casualties the company managed to achieve its objective. After capturing the hill, Manekshaw was hit by a burst of Light Machine Gun fire and was severely wounded in the stomach. Observing the battle, Major General David Cowan, the then commander of the 17th Infantry Division, spotted Manekshaw holding on to life and, having witnessed his valour in the face of stiff resistance, rushed over to him. Fearing that Manekshaw would die, the general pinned his own Military Cross ribbon to Manekshaw saying, "A dead person cannot be awarded a Military Cross. The official recommendation for the MC states that the success of the attack was largely due to the excellent leadership and bearing of Captain Manekshaw. This award was made official with the publication of the notification in a supplement to the London Gazette on 21 April 1942 (dated 23 April 1942).

9. **Indo-Pakistan War of 1971.** Under Manekshaw's direction, the army launched several preparatory operations in East Pakistan including training and equipping the Mukti Bahini (a local group of freedom fighters), and about three brigades from the regular Bangladesh troops were trained. As an additional measure, 75,000 guerrillas were trained and equipped with arms and ammunition. These forces were used to harass the Pakistani army stationed in East Pakistan sporadically in the lead up to the war.

10. The war started on 3 December 1971, when Pakistani aircraft bombed Indian Air Force bases in the western sector. Manekshaw instructed Lt Gen J F R Jacob, Chief of Staff Eastern Command, to inform the Indian prime minister that orders were being issued for the movement of troops from Eastern Command. The following day, the navy and the air force also initiated full-scale operations on both eastern and western fronts. The veto used by the Russians against the United States' proposal to implement a cease-fire in the United Nations proved decisive in securing India's victory. Manekshaw addressed the Pakistani troops three times via radio messages on the subject of surrender, assuring them that they would receive honourable treatment from the Indian troops. The messages were broadcast on the 9th, 11th and 15 December. The last two messages were delivered as replies to the messages from Maj Gen Rao Farman Ali and Lt Gen Amir Abdullah Khan Niazi. The Instrument of Surrender was formally signed on 16 December 1971.

11. **Promotion to Field Marshal.** After the end of the war, Indira Gandhi decided to promote Manekshaw to the rank of Field Marshal. Though Manekshaw was to retire in June 1972, his term was extended by a period of six months. On 3 December 1973, Manekshaw was conferred with the rank of Field Marshal at a ceremony held at Rashtrapati Bhavan.

MARSHAL OF THE AIR FORCE: ARJAN SINGH

12. **Early Life and Career.** Arjan Singh was born on 15 April 1919 in Lyallpur in the Punjab in what was then British India. He was educated at Montgomery, India (now in Pakistan). He entered the RAF College Cranwell in 1938 and was commissioned as a Pilot Officer in December 1939. As a distinguished graduate of the RAF College, Singh's portrait is found on the walls of the College's west staircase. Air Marshal Arjan Singh led No.1 Squadron, Indian Air Force into command during the Arakan Campaign in 1944. He was awarded the Distinguished Flying Cross (DFC) in 1944, and commanded the Indian Air Force Exhibition Flight in 1945.

13. **Career and Commands Held.** He was Chief of the Air Staff (CAS), from 1 August 1964 to 15 July 1969, and was awarded the Padma Vibhushan in 1965. He also became the first Air Chief Marshal of the Indian Air Force when, in recognition of the Air Force contribution in the 1965 war, the rank of the Chief of Air Staff was upgraded to that of Air Chief Marshal. After he retired in 1969 at the age of 50, he was appointed the Indian Ambassador to Switzerland in 1971. He concurrently served as the Ambassador to the Vatican. He was appointed High Commissioner to Kenya in 1974. He was member of the Minorities Commission, Government of India from 1975-1981. He was Lt. Governor of Delhi from Dec 1989 - Dec 1990 and was made Marshal of the Air Force in January, 2002. He expired on 16th Sep 2017.

CONCLUSION

14. It is mandatory for its civilian and military leaders to learn from military history to be ready to meet future challenges. We can learn a lot from the biographies of Field Marshal Cariappa, Field Marshal Sam Manekshaw and Marshal of the Air Force Arjan Singh. Therefore, it is important for all cadets to study the biographies of Field Marshal Cariappa and Field Marshal Sam Manekshaw. The Leadership qualities displayed by them worth emulating in every walks of life.

CHAPTER II : INDIAN ARMY WAR HEROES PVC

Introduction

1. Our Army has been involved in five major wars with our neighbouring countries. We have fought four wars with Pakistan and one with China. Other major operations undertaken by the army include Operation Meghdoot and Operation Cactus, Operation Pawan and Operation Rakshak. Apart from above mentioned operations, the army has also been an active participant in United Nations peacekeeping missions. The Indian Army has shown thorough professionalism, dedication and devotion while participating in all these wars and operations

RECIPIENTS OF THE PARAMVIR CHAKRA.

2. The country has awarded 21 Param Veer Chakras for exemplary courage and bravery to the following brave soldiers:-

- (a) Maj Somnath Sharma, 4 KUMAON Badgam, Kashmir November 3, 1947.
- (b) Naik Jadunath Singh Rajput Taindhara, Naushera, Kashmir February 6, 1948.
- (c) 2nd Lt Rama Raghoba Rane Bombay Engineers Naushera Rajouri Road April 8-11, 1948.
- (d) Company Havildar Major Piru Singh 6 Rajputana Rifles Tithwal sector, J&K July 18, 1948.
- (e) Lance Naik Karam Singh 1 Sikh Tithwal sector, J&K October 13, 1948.
- (f) Captain Gurbachan Singh Salaria 3/1 Gorkha Rifles Elizabethville, Katanga, Congo December 5, 1961.
- (g) Major Dhan Singh Thapa 1/8 Gorkha Rifles Ladakh October 21, 1962.
- (h) Subedar Joginder Singh 1 Sikh Tawang, Arunachal October 20-23, 1962.
- (i) Major Shaitan Singh 13 Kumaon Regiment Rezang La, Chusul sector, Ladakh November 18, 1962.
- (j) Lt Col A B Tarapore 17 Poona Horse Phillora, Sialkot, Pakistan September 11-16, 1965.
- (k) Company Quartermaster Havildar Abdul Hamid 4 Grenadiers Cheema, Khem Karan September 10, 1965.
- (l) Lance Naik Albert Ekka 14 Guards, Gangasagar, 6 1/2km west of Agartala December 4, 1971.
- (m) Flying Officer Nirmaljit Singh Sekhon No.18 "Flying Bullets" Squadron Srinagar December 14, 1971.
- (n) Second Lt Arun Khetarpal 17 Poona Horse, 47 Infantry Brigade Shakargarh sector December 16, 1971.
- (o) Major Hoshiar Singh Grenadiers Regiment Shakargarh sector December 17, 1971.
- (p) Naib Subedar Bana Singh J&K Light Infantry Siachen 1987.
- (q) Major Ramaswamy Parameswaram Mahar Regiment Sri Lanka (Operation Pawan) November 25, 1987.
- (r) Captain Vikram Batra 13 J&K Rifles July 7, 1999.
- (s) Lt Manoj Kumar Pandey 1/11 Gorkha Rifles July 2-3, 1999.
- (t) Grenadier Yogendra Singh Yadav 18 Grenadiers July 3-4, 1999.
- (u) Rifleman Sanjay Kumar 13 J&K Rifles July 4, 1999.

MAJ SHAITAN SINGH,PVC

3. **Introduction** Major Shaitan Singh was born on December 1, 1924 at Jodhpur in Rajasthan. His father was Lt Col Hem Singh Bhati.

4. **Military Action.** The 'C' Company of the battalion, led by Singh, held this crucial position at Rezang La, at a height of 5000 metres. The company area was defended by three platoon positions and the surrounding terrain isolated it from the rest of the battalion. The expected Chinese attack on Rezang La came on November 18 in the morning. It was the end of a very cold winter night, with light snow falling. The icy winds howling through Rezang La were biting and numbing. In the dim light of the morning, the Chinese were seen advancing through nullahs to attack No.7 and No.8 Shaitan Singh 2 platoon positions.

5. At 0500 hours when the visibility improved, both platoons opened up on the advancing Chinese with rifles, light machine guns, grenades and mortars. Indian artillery could, however, not be used. The nullahs were littered with dead bodies. The Chinese, though they failed the first frontal attack, were not discouraged. They subjected the Indian positions to intense artillery and mortar fire at about 0540 hours. Soon about 350 Chinese troops commenced advance through the nullahs. This time, No.9 Platoon, which held fire till the enemy was within 90 metres opened up with all weapons in their possession. Within minutes, the nullahs were again full of dead bodies, mainly of the Chinese. In frontal attack, the enemy, approximately 400 strong,

then attacked from the rear of the company position. An assault group of 120 Chinese also charged No.7 Platoon position from the rear. However, Indian Army 3-inch mortar killed many of them. When 20 survivors charged the post, about a dozen KUMAONIS rushed out of their trenches to engage them in a hand-to-hand combat.

6. Meanwhile, the Chinese brought up fresh reinforcements. The encirclement of No.7 Platoon was now complete. The platoon, however, fought valiantly till there was no survivor. No.8 Platoon also fought bravely to the last round. The Chinese announced a unilateral ceasefire on November 21, 1962. In this action, 109 KUMAONIS out of a total of 123 were killed. Of the 14 survivors, 9 were severely injured.

7. The Chinese suffered more than a thousand casualties. After the war was over, the body of Shaitan Singh was found at the same place, dead from the bullet wound and the freezing cold. Major Shaitan Singh was awarded Param Veer Chakra, the highest wartime gallantry medal, posthumously, for his leadership and devotion to duty.

CAPTAIN VIKRAM BATRA, PVC

8. **Early Life and Career.** Vikram Batra was born on 9 September 1974 in Ghuggar village near Palampur, Himachal Pradesh, to GL Batra and Jai Kamal Batra. He got his primary education from his mother, who herself was a teacher. He received his education up to Middle Standard at the D.A.V. Public School in Palampur and up to senior secondary stage in Central School, Palampur. After passing his 10+2 in 1992 from Central School Palampur, he got admitted in D.A.V. College, Chandigarh in B.Sc where he was adjudged the best N.C.C. Cadet (Air Wing) in two zones. Later, he was selected to join the Indian Military Academy in Dehradun in 1996 in Jessor company of Manekshaw Battalion, and was commissioned in the Indian Army as a Lieutenant of the 13 Jammu & Kashmir Rifles at Sopore, in Jammu and Kashmir. He rose to the rank of Captain.

Heroic Act of Capt Bikram Batra, PVC in Kargil War

9. During the Kargil invasion of 1999 by Pakistan, Lt Batra (at time), 13 JAK Rifles, and his Delta Company were ordered to recapture peak 5140 on June 19, 1999 five weeks after the war began. Nicknamed *SherShah* ('Lion King') in Urdu for his courage which also doubled as his call sign, he decided to approach the hill from the rear, aiming to surprise the Pakistani defenders. He and his men ascended the sheer rock-cliff, but as the group neared the top, the enemy pinned them on the face of the bare cliff with machine gun fire. Captain Batra, along with five of his men, climbed up regardless and after reaching the top, hurled two grenades at the machine gun post. He single-handedly killed three enemy soldiers in close combat. He was seriously injured in the process, but insisted on regrouping his men to continue with the mission. Inspired by the courage displayed by Captain Batra, the soldiers of 13 JAK Rifles charged the enemy position and captured Point 5140 at 3:30 a.m. on 20 June 1999. His company is credited with killing at least eight Pakistani intruders and recovering a heavy machine gun.

20. The capture of Point 5140 set in motion a string of successes, such as Point 5100, Point 4700, Junction Peak and Three Pimples. Along with fellow Captain Anuj Nayyar, Batra led his men to victory with the recapture of Point 4750 and Point 4875. This led to the fall of Tiger Hill and India's eventual hold on the valley was strengthened.

21. Nine days later, Vikram Batra was assigned to an urgent mission to recapture peak 4875. This was one of the most difficult peaks to capture as the Pakistani troops sat above the peak at 16,000 feet and the climb gradient was 80 degrees. The fog made matters worse for Batra and his team. In the early morning hours of 7 July 1999, he commanded a mission to rescue an injured officer during a Pakistani counterattack against Point 4875. During the rescue attempt, he pushed aside his Subedar, saying "*Tu baal-bacche dar hai, hat ja peeche.*" (*You have children, step aside*) and was killed in action while clearing enemy positions. His last words were, "*Jai Mata Di.*", which is a Punjabi creed referring to Durgadevi, the Hindu Goddess of Victory.

CONCLUSION

22. Soldiers face the dangers and vagaries of war and sacrifice their lives for their motherland. The Indian Nation also honours its bravest of the brave soldiers by conferring on them the highest gallantry award "Param Vir Chakra" as recognition of their bravery and sacrifice.

CHAPTER III : STUDY OF BATTLES OF INDO-PAK WAR 1965, 1971, AND KARGIL

Introduction

1 The partition of the subcontinent came into effect on 15 August 1947, when India gained independence. Pakistan declared independence a day earlier. At the time of independence the old Indian Army stood divided between Pakistan and India. Instead of large scale celebrations, riots and mass killing between Hindus and Muslims in Punjab and Bengal intensified. It also led to acute suffering and misery of the displaced people, a part from colossal loss of precious human lives and destruction of property due to communal riots and retribution.

2 Taking advantage of communal strife, in Oct 1947 Pakistani troops soon crossed over into Kashmir to participate into an undeclared war with India. The tribal volunteers along with Pakistani regulars had by then overrun large tracts of Jammu province and the Valley, which shared a porous border with Pakistan. It was when they had reached Badgaon, on the suburbs of Srinagar that the Maharaja of J&K signed the Instrument of Accession and put in a bid for India's military assistance. Indian Army then swung into action to save J&K.

INDO-PAKISTANI WAR 1965

3 The Indo-Pakistan war of 1965 was a culmination of skirmishes that took place between April 1965 and September 1965 between Pakistan and India. The conflict began following Pakistan's Operation Gibraltar, which was designed to infiltrate forces into Jammu and Kashmir to participate in insurgency against Indian rule. In retaliation, India reacted swiftly and launched a counter attack and a second confrontation with Pakistan took place in 1965, largely over Kashmir. Pakistani President Ayub Khan launched Operation Gibraltar in August 1965, during which several Pakistani paramilitary troops infiltrated into Indian-administered Kashmir and attempted to ignite an anti-India agitation in Jammu and Kashmir. Pakistani leaders believed that India, which was still recovering from the disastrous Sino-Indian War, would be unable to deal with a military thrust and a Kashmiri rebellion. Pakistan launched Operation Grand Slam on 1 September, invading India's Chamb-Jaurian sector.

4 **Offensive on Pakistan.** Initially, the Indian Army met with considerable resistance in the northern sector. After launching prolonged artillery barrages against Pakistan, India was able to capture three important mountain positions in Kashmir. By 9 September, the Indian Army had made considerable inroads into Pakistan. India had its largest haul of Pakistani tanks when the offensive of Pakistan's 1st Armoured Division was blunted at the Battle of Asal Uttar, which took place on 10 September near Khemkaran in Punjab. Another tank battle of the war came in the form of the Battle of Chawinda, the largest tank battle in history after World War II. Pakistan's defeat at the Battle of Asal Uttar and Dograi hastened the end of the conflict.

5 **Battle of Dograi.** To relieve pressure at Chamb-Jaurian Sector in J&K, 15 Infantry Division launched offensive in Lahore sector. 3 JAT of 54 Infantry Brigade crossed Ichogil canal and captured the town ship of Dograi and was just 13 miles from Lahore on 23 September 1965. On 23 September 1965 cease fire was announced. A decision to return back to pre-war positions was taken following the Tashkent Declaration.

INDO-PAKISTAN WAR 1971

6 An independence movement broke out in East Pakistan which was brutally crushed by Pakistani forces. Due to large-scale atrocities against them, thousands of Bengalis took refuge in neighbouring India causing major refugee crisis. In early 1971, India declared its full support for the Bengali rebels, known as Mukti Bahini, and Indian agents were extensively involved in covert operations to aid them. Wary of India's growing involvement in the Bengali rebellion, the Pakistan Air Force (PAF) launched a preemptive strike on 10 Indian air bases at Srinagar, Jammu, Pathankot, Amritsar, Agra, Adampur, Jodhpur, Jaisalmer, Uttarlai and Sirsa at 1745 hours on 3 December. This aerial offensive, however, failed to accomplish its stated objectives and gave India its excuse to declare a full-scale war against Pakistan the same day. By midnight, the Indian Army, accompanied by Indian Air Force, launched a major three-pronged assault into East Pakistan. The Indian Army won several battles on the eastern front including the decisive Battle of Hilli, which was the only front where the Pakistani Army was able to build up considerable resistance. India's massive early gains were largely attributed to the speed and flexibility with which Indian armoured divisions moved across East Pakistan.

7 **Battle of Longewala.** Pakistan launched a counter-attack against India on the western front. On 4 December 1971, 'A' company of the 23rd Battalion of India's Punjab Regiment detected and intercepted the movement of the 51st Infantry Brigade of the Pakistani Army near Ramgarh, Rajasthan. The battle of Longewala ensued during which the A company, though being outnumbered, thwarted the Pakistani advance until the Indian Air Force directed its fighters to engage the Pakistani tanks. By the time the battle had ended,

38 Pakistani tanks and 100 armoured vehicles were either destroyed or abandoned. About 200 Pakistani troops were killed in action during the battle while only 2 Indian soldiers lost their lives. Pakistan suffered another major defeat on the western front during the Battle of Basantar which was fought from 4 December to 16 December. By the end of the battle, about 66 Pakistani tanks were destroyed and 40 more were captured. In return, Pakistani forces were able to destroy only 11 Indian tanks. By 16 December, Pakistan had lost sizeable territory on both eastern and western fronts.

8 **Surrender of Pakistan Army in Dhaka.** Under the command of Lt General JS Arora, the '3 Corps' of the Indian Army, which had invaded East Pakistan, entered Dhaka and forced Pakistani forces to surrender on 16 Dec 1971, one day after the Battle of Basantar. After Pakistan's Lt General AAK. Niazi signed the Instrument of Surrender, India took more than 90,000 Pakistani prisoners of war.

KARGIL CONFLICT- 1999

9 In 1998, India carried out nuclear tests and a few days later, Pakistan responded by more nuclear tests giving both countries nuclear deterrence capability, although India had exploded three hydrogen bombs which Pakistan lacks. Diplomatic tension ceased after the Lahore Summit was held in 1999. The sense of optimism was short-lived, however, since mid-1999 Pakistani paramilitary forces and Kashmiri insurgents captured deserted, but strategic, Himalayan heights in the Kargil district of India. These had been vacated by the Indian army during the onset of the inhospitable winter and were supposed to reoccupy in spring. Once the scale of the Pakistani incursion was realised, the Indian Army quickly mobilized about 200,000 troops and Operation Vijay was launched.

10 **Battle of Tololing.** The Battle of Tololing, was one of the pivotal battles in the Kargil war between Indian Armed forces and troops from Northern Light Infantry who were aided by other Pakistan irregulars in 1999. Tololing is a dominating feature overlooking Srinagar-Leh (NH1D) and was a vital link. The terrain was such that frontal attacks had to be launched which resulted in heavy casualties. The three week assault finally culminated with India taking control of the peak and changing the course of the war. 23 Indian soldiers were killed in the final assault, resulting in one of the costliest battles of the entire war. Other assaults, slowly tilted the combat in India's favour. Nevertheless, some of the posts put up stiff resistance, including Tiger Hill (Point 5140) that fell only later in the war.

11 The Indian Army mounted some direct frontal ground assaults which were slow and took a heavy toll given the steep ascent that had to be made on peaks as high as 18,000 feet (5,500m). Two months into the conflict, Indian troops had slowly retaken most of the ridges they had lost; according to official count, an estimated 75%–80% of the intruded area and nearly all high ground was back under Indian control. On 4 July 1999, Pakistan's Prime Minister Sharif agreed to withdraw Pakistani troops under US pressure and the fighting came to a gradual halt, but some Pakistani forces remained in positions on the Indian side of the LOC.

12 The Indian Army launched its final attacks in the last week of July; as soon as the Drass sub sector had been cleared of Pakistani forces, the fighting ceased on 26 July. 26 July has since been marked as 'Kargil Vijay Diwas' (Kargil Victory Day) in India. By the end of the war, India had resumed control of all territory south and east of the Line of Control, as was established in July 1972 as per the Shimla Accord.

CONCLUSION

13 Wars with Pakistan have been a regular affair in the long history of both countries along with conflicts with China. This is a compulsive and existential necessity for the Pakistani State controlled by their military. The sacrifices made by the Indian Army to safeguard the sovereignty and integrity of the nation are great, for which the whole nation is proud of its achievements.

CHAPTER IV: WAR MOVIES

1 **Introduction.** Battles are fought in the wilderness and away from limelight, however ferocious and destructive they may be. Historians have always tried to piece together those little but significant actions by individuals and groups bringing out their valour and courage for the consumption of future generations. Here we will see three visual documentaries on Indo – Pak wars of 1965, 1971 and 1999 to gain further insight on the subject.

Documentary on Indo – Pak War 1965

2 Screening of documentary on Indo – Pak war 1965.

Documentary on Indo – Pak War 1971

3 Screening of documentary on Indo – Pak war 1971.

Documentary on Kargil Conflict 1999

4 Screening of documentary on Indo – Pak Kargil Conflict, 1999.

CONCLUSION

5 The study of the Indo-Pak War brings out the sacrifices made by the Armed Forces of India. It is heartening to see the role of Armed Forces in maintaining the sovereignty and integrity of our nation. One must salute the war heroes and all those who have laid down their lives for the motherland.

SUMMARY

6 Field Marshal KM Cariappa, OBE was the first native Indian Chief of Army Staff of the Indian Army. He is only among three Indian Army officers to hold the highest rank of Field Marshal. He was appointed as the Commander-in-Chief of the Indian Military in 1949, received the rank of Field Marshal On 14th January 1986 at the age of 87.

7 The Param Vir Chakra (PVC) is India's highest military decoration award for highest degree of valour or self-sacrifice in the presence of enemy. The PVC was established on 26 January 1950 by the President of India. The medal was designed by Mrs Savitri Khanolkar.

8 Maj Shaitan Singh, PVC was awarded ParamVir Chakra, the highest wartime gallantry medal, posthumously, for his outstanding leadership during 'Battle of Rejang La'.

9 Captain Vikram Batra, PVC was posthumously awarded with the ParamVir Chakra for his actions during the 1999 Kargil Conflict. He led one of the toughest operations in mountain warfare in Indian history.

10 India has fought four wars with Pakistan so far, 1947-48 War, 1965 War, 1971 War and Kargil Conflict

11 War of 1965.

- (a) Pakistan launched Operation Gibraltar to support insurgency in J&K against Indian rule.
- (b) In retaliation India launched Operation Grand Slam in western sector against Pakistan.
- (c) The famous battles of 1965 war include Battle of Asal-Uttar, Battle of Dograi etc.
- (d) Battle of Chawinda was the largest tank battle in the history after World War II.

12 **War of 1971.**

- (a) In early 1971, India declared its full-support for the Bengali rebels, Known as Mukti Bahini.
- (b) Battle of Longewala was fought on 4 December 1971.
- (c) On 16 December 1971, more than 90000 Pakistani soldiers surrendered at Dhaka.

13 Kargil Conflict 1999 was fought on the heights of Kargil and Drass in 1999. In mid-1999 Pakistani paramilitary forces and Kashmiri insurgents captured deserted, but strategic, Himalayan heights in the Kargil district of India. Indian forces painstakingly recaptured all posts one after another in a bloody conflict. Operation Vijay was declared success on 26 July 1999.

CHAPTER I : INTRODUCTION TO COMMUNICATION AND LATEST TRENDS

Introduction

1. Communication is the easy transmission of thoughts and ideas from one individual to another and vice-a-versa and reception and understanding of others ideas in the original form. Various forms of communications are sign language, voice, written script, line transmission, radio wave, space wave and highly complex digital communication. In modern battlefield, activities of the army have become very complex and require frequent communication between commanders and troops to achieve the desired results. Importance of communications is becoming a major ingredient of modern warfare.

2. **Line Communication** The invention of telephone by Graham Bell revolutionized the world of communications as individuals were able to speak directly to each other. This is the basic means of signal communications for a force which is static. A telephone is by far the best means of signal communication between individuals following are advantages and disadvantages given below: -

(a) **Advantages.**

- (i) Reliable and practically free from electrical interference.
- (ii) Relatively secure.
- (iii) Number of circuits and message carrying capacity is more but limited only by availability of material and manpower.

(b) **Disadvantages.**

- (iv) Vulnerable to physical interference and enemy interception along the entire length of the route.
- (v) Takes time to construct.
- (vi) Inflexible once it is laid.
- (vii) Expensive in men and material

RADIO COMMUNICATION

3. **Radio Communication.** Radio is the technology of using radio waves to carry information such as sound, by systematically modulating properties of electromagnetic energy waves. The information in the waves can be extracted when radio waves strike an electrical conductor and transformed back into its original form. Radio communication requires the use of both transmitting and receiving equipment. The transmitting equipment, which includes a radio transmitter and transmitting antenna, is installed at the point from which messages are transmitted. The receiving equipment, which includes a radio receiver and receiving antenna, is installed at the point at which messages are received

4. **Wave.** A wave can be described as a disturbance that travels through a medium from one location to another location. Types of waves can broadly be divided in two parts :-

(b) **Mechanical Waves.** A mechanical wave is a wave that is an oscillation of matter, and therefore transfers energy through a medium.

(c) **Electromagnetic Waves.** Electromagnetic waves are waves which can travel through the vacuum of outer space.

(d) **Propagation of Wave.** The mode of propagation of electromagnetic waves (EMW) from transmitter to receiver depends upon the frequency employed. These can be of following types :-

- (i) Sky Wave Propagation.
- (ii) Space Wave Propagation
- (iii) Ground Wave Propagation.
- (iv) Tropospheric Scatter.

5. Radio Communication involves Net Radio and Radio Relay.

Net Radio

6. Net radio is the basic means of signal communication for any mobile force. Efficiency of net radio communication is appreciably affected by factors such as weather, terrain, power output of the set, state of training of operators and equipment maintenance. It provides facilities for the following :-

- (a) **Radio Telephony** – Simple one way voice communication, depending on the type of equipment available.
- (b) **Radio telegraphy** for transmission of message and key conversations.
- (c) **Tele printers** over radio transmission.

7. **Advantages.**

- (a) Vulnerable only at terminal and is therefore reasonably protected from enemy action except by a direct hit.
- (b) Flexible hence can be rapidly re-arranged in the event of regrouping.
- (c) Rapid in establishing communication.
- (d) Works on the move although range obtained will be much less than when stationary.
- (e) Economical in personnel and equipment.

8. **Disadvantages.**

- (a) Inherently insecure and susceptible to enemy interception which necessitates the use of codes and ciphers with a consequent delay in clearing traffic and overall increase in operating personnel.
- (b) Net radio being inherently insecure demands a considerable degree of security consciousness on the part of the users. This means adherence to standard procedure and security codes.

Radio Relay

9. Radio relay implies that a series of radio transmitters and receivers normally spaced between 20-35 Kms apart and are used to provide point signal communication.

10. **Advantages.**

- (a) Replace line with considerable economy of manpower and stores.
- (b) It can be operated over area where for reasons of ground or enemy activity use of line may not be possible.
- (c) Provides greater flexibility than line.
- (d) Quick to set up and move except in mountainous country.
- (e) By its ability to employ multichannel equipment radio relay provides more tele-printer circuits over one link than can normally be provided over the average field cable. Thus it has much greater traffic handling capacity.

11. **Disadvantages.**

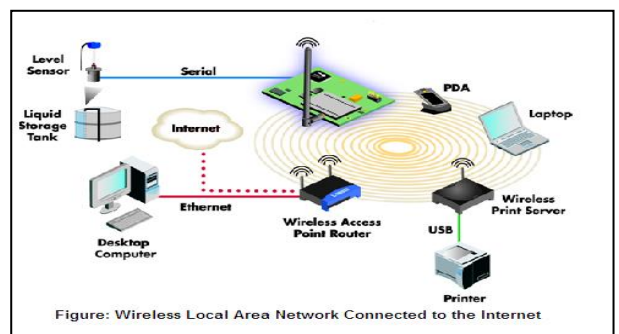
- (a) Liable to interception and hence insecure. Has relatively greater security than net radio, depending upon the sitting and direction of the beams.
- (b) Liable to interference from enemy jamming although not as much as in the case of net radio.
- (c) Terrain between stations must be reasonably suitable to get a 'quasi optical path', this presents difficulty in sitting.
- (d) Location of terminal and intermediate stations may not suit tactical layout and may, therefore, create additional protection requirements.
- (e) It cannot work on the move.
- (f) Slightly more expensive in men and material than in the case of net radio.
- (g) Needs critical sitting.

CHARACTERISTICS OF WIRELESS TECHNOLOGY (MOBILE, WI-FI ETC)

12. **Features of Wi-Fi Technology.** Wireless operations permit services, such as a long-range communications, which are impossible or impractical to implement with the use of wires. Information is transferred in this manner over both short and long distances.

13. The following list summarizes some of the benefits of a Wi-Fi network

- (a) **Unmatched Mobility and Elasticity.**



Wi-Fi, is allowing the new intensity of connectivity without giving up functions. Wi-Fi introduced various types of utilities such music streamers that transmit your music to speakers without any wire you can also play music from the remote computer or any other attached to the network. The most important now you can play online radio. Wifi technology system is rather remarkable, you can download songs, send an email and transfer files expediently at sky-scraping speed and you can move your computer easily because your WiFi network has no cable to disrupt your work so we can say that it is quite easy, helpful and most of all expedient.

(b) **Fortress Technology** Wi-Fi providing secure wireless solutions support the growth and release of a prototype mobile ad hoc wireless network for use in the wireless strategic skirmish.

(c) **Support an Entire Age Bracket.** Wi-Fi technology has several advantages, it supports an entire age bracket and create as connection between components on the same network and have the ability to transfer data between the devices and enable different kinds of devices such as game, MP3 player, PDA's and much more!

(d) **Convenient and Available Everywhere.** Wi-Fi is a convenient technology and where the network range station exists you are online during travel; you can equip with a Wi-Fi network and get connected. You will automatically connect to the internet if you are near a hotspot. These days WiFi exist everywhere with all its wonders.

(e) **Faster and Secure** With Wi-Fi, you can get a high speed of internet because it is very fast than DSL and Cable connection you can establish a Wi-Fi network in small space now you don't need any professional installation just connect to a power outlet with an Ethernet cord, and start browsing. Wi-Fi security system for Threats makes it more renewable and its tool protect your VPN and secure web page. You can easily configure the device to take better performance. The standard devices, embedded systems, and network security make it more powerful.

(f) **Wi-Fi with no Limitation** You can use a "Wi-Fi" network with no limitation because it can connect you worldwide. You can easily cater to your requirements with Wi-Fi networking applications because the power consumption is not very high as compared to another bandwidth networks. The future of wireless internet network communications is bright.

(g) **Extension of Wi-Fi Technology** It is because of this wireless technology that so many other advancements could take place. Have you even thought of your TV to be supported by WI-FI? Well if you didn't, start thinking now. There are smart TV's in a market that connects to the internet. Having the internet on TV makes it possible to watch you tube videos, Net Flix and so much more. Read more about Wi-Fi TV technology on World Wide Web for more information

(h) **Cost Reduction.** As mentioned above, the absence of wires and cables brings down cost.

(i) **Flexibility.** Extended access, cost reductions, and mobility create opportunities for new applications as well as the possibility of creative new solutions for legacy applications.

TERMINAL EQUIPMENT AND LIMITATIONS OF WI-FI

14. **Mobile Telephones.** One of the best-known examples of wireless technology is the mobile phone, also known as a cellular phone, with more than 4.6 billion mobile cellular subscriptions worldwide as of the end of 2010. These wireless phones use radio waves from signal-transmission towers to enable their users to make phone calls from many locations worldwide.

15. **Wireless Data Communications.** Wireless data communications are an essential component of mobile computing. The various available technologies differ in local availability, coverage range and performance, and in some circumstances, users must be able to employ multiple connection types and switch between them.

16. Wi-Fi technology is not perfect and has many flaws that limit its use which are as follows:

(a) **Security.** Because wireless transmissions can pass through walls, security is an issue.



- (b) **Wireless Reception.** Varies from area to area, even within your own apartment. It's not always guaranteed that you'll have a connection to the Internet.
- (c) **Interference.** Call quality is greatly influenced by the environment, is particularly sensitive to electromagnetic radiation generated by other household appliances.
- (d) **Compatibility Issue.** Despite the global standardization, many devices from different manufacturers are not fully compatible, which in turn affects the speed of communication

CHARACTERISTICS OF WALKIE / TALKIE

17. A Walkie/Talkie (formally known as a hand held transceiver) are a hand-held, portable, two-way radio transceiver. Its development during the Second World War is credited to Donald L. Hings, radio engineer Alfred J. Gross, and engineering teams at Motorola. Where a phone's ear piece is only loud enough to be heard by the user, a walkie-talkie's built-in speaker can be heard by the user and those in the user's immediate vicinity. Hand-held transceivers may be used to communicate between each other, or to vehicle-mounted or base station.

18. **Radio set GP338 Motorola.** Defence organizations use hand held radios for a variety of purposes. Radio Set Gp338 Motorola can communicate on a variety of bands and modulation schemes.

19. **Facilities / Features**

- (a) **Facilities.**
 - (i) It is portable and light in weight.
 - (ii) Can be operated easily.
 - (iii) It can be operated in VHF/UHF and 2 way simplex mode.
 - (iv) 128 channel of this radio set can be preset into 8 zone.
 - (v) Option of selective call facilities available.
 - (vi) Call alert can be given to receiver station.
 - (vii) Can select required zone.
 - (viii) Start/stop facilities of scan operation available.
 - (ix) Add/Delete from scan list facilities available.
- (b) **Weight and Measure.** Weight and measure of this radio set is as follows:
 - (i) Dimension with Nickel Metal-Hydride high capacity battery-137x57.5x37.5mm.
 - (ii) Weight with Nickel Metal-Hydride high capacity battery-420gm.
- (c) **Frequency Range.** Frequency of this radio set is as under:-
 - (i) 136 MHz to 174 MHz on VHF mode.
 - (ii) 403 MHz to 470 MHz on UHF mode.
- (d) **Communication Range.** Communication range of this radio set is 4 to 5 Km and 20 to 40 Km with repeater.
- (e) **Power Supply.** Power supply can be provided to this radio set by the high capacity 7.2 volt batteries.
- (f) **Battery Charging Time.** 1 hour for high capacity 7.2 volt batteries.
 - (i) Power output (HF) 1 to 5W att.
 - (ii) Power output (UHF) 1 to 4 Watt.

20. **WALKIE/TALKIE.** This radio set has been divided into four portions.

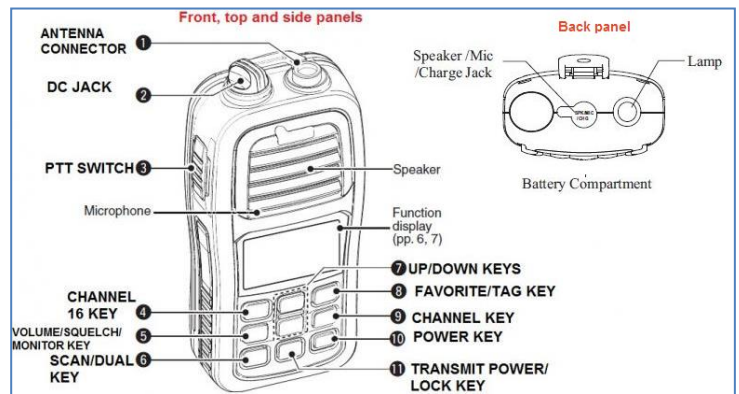


(a) **Top Panel.** Contains On/Off Volume Knob, Channel Selector Knob, Top Button and Antenna

(b) **Side Button.** Name and functions of the large control on side panel is Press to Talk Switch and Right Side Accessory Mount.

(c) **Front Panel.** There are a total of six buttons i.e. Exit Key, Up Key, and Menu Select Key. Front Panel Key Pad. This is an alphanumeric keypad. There are 10 keys on this pad from 0 to 9. One key star and one extract key. LCD Display. This is a 14-character LCD display window and 14 types of indicators are displayed.

(d) **Back Panel.** The battery is fixed on this back panel of radio.



LATEST TRENDS AND DEVELOPMENT IN COMMUNICATION

Tropo Scatter, Modem, Fax, and Telex

21. **Tropo scatter.** The lower layer of the atmosphere below 15 Km height is called tropospheric region. Communication carried out in this layer use the principle of tropo scatter. In this system micro waves are transmitted in the Ultra High Frequency (UHF) and Super High Frequency (SHF) band to achieve Radio Communication over the horizon covering a range between 70 Km to 1000 Km.



22. **MODEM (Modular-Demodulator).** This device is used to convert computer generated output (Digital signals) that can be transmitted on a telephone line. Modems are required at both the sending and receiving computers.

23. **Fax.** This is common short form of FACSIMILE which is one of the memory type electronic mail and message systems with the following advantages:-

- (a) Can transmit graphics as well as Alpha numeric information (letters and numbers).
- (b) Reduce time and eliminates transmission error.
- (c) Use any transmission medium eg. Telephone, line, micro radio wave.



24. **Telex.** This is the abbreviated form of TELE PRINTER EXCHANGE. As cable is used in this type of communication devices to connect two such instruments it restricts its range of operation.

(a) **Advantage.**

- (i) Re-generative repeaters in a network can increase range; however voice signal cannot be re-generated.
- (ii) Can be used over a telephone network.
- (iii) Can receive messages when unattended.
- (iv) Message is recorded in a printer form.

(b) **Disadvantages.**

- (i) The Equipment is costlier than a telephone set.
- (ii) Key in error due to the need for a human operator to send-receive message.
- (iii) Lack of privacy since any one can read the printed output.



25. **Satellite.** Satellites are used for many purposes. Common types include military and civilian Earth observation satellites, communications satellites, navigation satellites, weather satellites, and space telescopes. Space stations and human spacecraft in orbit are also satellites. Satellite orbits vary greatly, depending on the purpose of the satellite, and are classified in a number of ways. Well-known (overlapping) classes include low Earth orbit, polar orbit, and geostationary orbit.

26. **Optical Fibre Communication Computer System.** Hollow tubes made of corning glass with an outer protective coating of rubber/plastic etc. are what constitute optical fibers. These fibers are very delicate and small in diameter.

(a) **Advantages.**

- (i) It has wide band width carrying different types of information from low speed voice signal to high speed computer data.
- (ii) Less power requirement.
- (iii) Small cable size.
- (iv) No electromagnetic interference.

(b) **Disadvantages.**

- (i) Expensive in terms of equipment and manufacturing.
- (ii) Requirement of experts to run the system

27. **Computer System.** Strictly speaking a computer is any calculating device. The name is derived from a Latin word "Computer" meaning to reckon or compute. However, the term computer has come to mean a special type of calculating machine having certain characteristics.

28. **Internet.** It provides an instant, trouble free and cheap means of communications. Internet is therefore a collection of individual data networks connected together in such a way that data can be exchanged back and forth between networks widely separated. Electronic Mail, Web- Browsing and Voice Mail are the main facilities of internet.

29. **Cell Phone.** Cellular radio network was first introduced in 1980. It provides a mobile subscriber access to the global telephone network. It is a rapidly expanding technology with high rates of obsolescence.

30. **Multimedia.** It is a computer technology that displays information using a combination of full motion video animation, sound graphics and text with high degree of user interaction.

31. **Video-Conferencing Systems.** These provide the full benefits of face to face communication with sound, graphics and simultaneous transmission of data. The system enables people widely separated geographically to inter- act without having to meet at one place.

32. **Videophone.** It is a system that enables us to transmit an image via digital tele network, making visual contact has been made possible over great distances, apart from transferring speech. Facilities provided by videophone are:

- (a) Can transmit speech as well as colour video.
- (b) Conduct of video conferences.

- (c) Called subscriber is seen on the monitor.
- (d) High quality of voice.
- (e) Speed of sending/ receiving can be adjusted by the user.
- (f) Map over-lays can be transmitted.

Information Technology

33. Information Technology or IT for short, refers to the creation, gathering, processing, storage, presentation and dissemination of information, and also the processes and devices that enable all this to be done. IT stands firmly on the hardware and software of a computer and the telecommunications infrastructure. What has made the big difference in recent years is not the fact that individual computers have dramatically improved in their capabilities, but that all those information islands are being connected by digital highways made possible through the use of the telecommunications infrastructure by the computers, which, largely explains why the internet and the WWW have begun to play such a significant role in our use of computers.

CONCLUSION

34. Today the technology has advanced in all spheres of life. Even the defence forces are well poised to exploit the state-of-art modern communication techniques for meeting the requirements of the Indian Army in the 21st century. Some of the areas where the army is already in the process of exploiting are the Cellular Radio (both GSM & CDMA), WLL, mobile trunked radio, mobile satellite systems, OFC and so on. Therefore, it is very important to be well versed with the latest trends in information technology to gain maximum advantage from the same



SUMMARY

- 35. Line. This is the basic means of signal communications for a force which is static.
- 36. Radio Communication. Radio communication involves Net Radio and Radio Relay. Net radio is the basic means of signal communication for any mobile force.
- 37. Radio Relay. Radio relay implies that a series of radio transmitters and receivers normally spaced between 20-35 Km apart and are used to provide point signal.
- 38. Wireless. Wireless operations permit services, such as a long-range communications, that are impossible or impractical to implement with the use of wires.
- 39. Mobile Telephones. These wireless phones use radio waves from signal-transmission towers to enable their users to make phone calls from many locations worldwide.
- 40. Wireless Data Communications. Wireless data communications are an essential component of mobile computing. The various available technologies differ in local availability, coverage range and performance.
- 41. Wi-Fi. Wi-Fi is a wireless local area network that enables portable computing devices to connect easily to the Internet. Wi-Fi has become the de facto standard for access in private homes, within offices, and at public hotspots. Cellular data service offers coverage within a range of 10-15 miles from the nearest cell site.
- 42. A walkie-talkie is a hand-held, portable, two-way radio transceiver.
- 43. Troposcatter is a system in which micro waves are transmitted in the UHF and SHF band to achieve radio communication over the horizon covering a range between 70 Km to 1000 Km.
- 44. Modem (modulator-demodulator) is a network hardware device that modulates one or more carrier wave signals to encode digital information for transmission and demodulates signals to decode the transmitted information.
- 45. A satellite is an artificial object which has been intentionally placed into orbit. Such object is sometimes called artificial satellite to distinguish them from natural satellites such as Earth's Moon.

46. Satellites are used for a large number of purposes. Common types include military and civilian earth observation satellites, communications satellites, navigation satellites, weather satellites, and research satellites.

47. Multimedia is content that uses a combination of different content forms such as text, audio, images, animation, video and interactive content. Multimedia contrasts with media that use only rudimentary computer displays such as text-only or traditional forms of printed or hand-produced material.

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